Pre-registered backtests of option and equity strategies on trade-sampled US market data, 2022–2024

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Abstract

We report pre-registered backtests of option and equity strategies on one dataset: US option trades with attached NBBO, end-of-day (EOD) option chains, equity quotes around the open and close, and auction prints for 18 large US equities and ETFs, from 2022-01-03 to 2024-12-31. Four rounds of research cover 25 strategy specifications. Each strategy was written down before its out-of-sample data was used, filled at a declared execution tier, and charged a realistic fee schedule. None clears a confirmatory bar, which we define as an out-of-sample result that is also statistically distinguishable from the best of many trials.

One lead remains. Delta-weighted option order flow predicts next-day stock returns. In the wave-2 2024H2 holdout (H4) the long-short book has Sharpe 3.26 and a Fama–MacBeth $t = 2.32$. In the wave-3 2024 test with realistic fees (S3, 1-day hold) the Sharpe ratio is 1.71. The development periods give Newey–West $t = 1.48$ (wave 2) and 1.38 (wave 3), with deflated Sharpe ratios of at most 0.17. The signal needs a 2025 sample before anyone trades it.

The option tape is a capped, trade-sampled record. It supports four-leg 0DTE SPXW fills on about 80% of days, but it almost never fills a two-leg straddle at a pre-chosen strike. Several textbook strategies are therefore not testable on this data, independent of their economics.

We also combine the 13 primary sleeves, with undeployed capital earning SOFR and weights estimated on 2022–2023 only. In 2024 a Bayes–Stein Markowitz book capped at 25% per sleeve earns 3.67% over SOFR (static weights; Sharpe 1.89), and 1/N earns 4.21% (Sharpe 2.00). Every sleeve's 2024 outcome was known when the allocation study was designed, so we treat these figures as the basis for a paper-trading proposal (Section 7), not as evidence.

1 Introduction and overview of potential strategies

This study asks which option and equity strategies survive realistic execution on a three-year sample of US market data. The data are described in Section 2. The research ran in four rounds ("waves"), and each later wave was designed after seeing the failures of the earlier ones (Section 3). We tested 25 strategy specifications in total. We describe them here by family, with the economic reason each family might work.

Variance-risk-premium harvesting. Index options carry a variance risk premium, meaning that option-implied variance exceeds subsequently realized variance on average (Carr and Wu, 2009; Bollerslev et al., 2009). Defined-risk short-premium structures, such as credit spreads, iron condors and iron flies, might collect part of that premium. Waves 0 and 1 tested EOD put and call credit spreads, condors and debit verticals on SPY, QQQ and SPX (Section 6.1). Wave 2 moved to same-day (0DTE) SPXW options, which may embed a premium paid by retail traders and

hedgers for convexity (Bandi et al., 2023; Beckmeyer et al., 2023; Vilkov, 2023) (H1, Section 6.3). Wave 3 added a 28-day SPXW iron fly timed by the VIX term structure (Johnson, 2017) (S5, Section 6.12) and straddles sorted on realized minus implied volatility (Goyal and Saretto, 2009) (S6, Section 6.13). A 0DTE pin test (S7, Section 6.14) asks whether hedging of large open interest pulls the index toward one strike (Ni et al., 2005).

Option-implied information. Informed traders use options for leverage and concealment, so option order flow and implied-volatility asymmetries can lead stock prices (Easley et al., 1998; Pan and Poteshman, 2006; Cremers and Weinbaum, 2010; Xing et al., 2010; Johnson and So, 2012). We measure this with delta-weighted signed option volume. The signal feeds a cross-sectional long-short stock book (H4 and S3, Section 6.6) and a pre-announcement test (S4, Section 6.11).

Earnings-event strategies. Implied volatility rises into scheduled earnings, and stocks earn higher returns in the days before them (Gao et al., 2018a; Frazzini and Lamont, 2007; Barber et al., 2013). After the announcement, prices may under-react to the news (Brandt et al., 2008; Chan et al., 1996). We test a pre-earnings straddle (H5, Section 6.7), a pre-earnings stock premium (S1, Section 6.9) and post-announcement drift (S2, Section 6.10).

Overnight, intraday and calendar effects. A large share of the equity premium may accrue overnight (Lou et al., 2019; Cliff et al., 2008), and delta-hedged option returns may change sign between night and day (Muravyev and Ni, 2020; Jones and Shemesh, 2018). Late-day order flow may push the day's move further into the close (Gao et al., 2018b; Baltussen et al., 2021). We test an overnight short straddle (H2, Section 6.4), an overnight equity drift (H3, Section 6.5), end-of-day momentum (H6, Section 6.8), daytime reversals (Akbas et al., 2022) (S8, Section 6.15), the pre-FOMC drift (Lucca and Moench, 2015) and the turn-of-the-month effect (Ariel, 1987; Lakonishok and Smidt, 1988) (S9 and S10, Section 6.16).

Design rules we learned. Three rules shaped the later waves.

1. *Cross the spread once, or not at all.* Spread crossing turned the documented overnight option premium negative (H2): the spread alone was about four times that premium. Designs that cross once and settle (H1, S5), or that trade only in auctions (H3, H4, S1–S3, S8–S10), remove most of that cost.
2. *Longer maturities cost less relative to premium.* At 8–30 days to expiry (DTE), crossing the spread once costs about 0.6% of premium, against about 8% for a 0DTE round trip (F1, Section 6.2).
3. *Breadth and sample length determine power.* With three years of data the standard error of an annualized Sharpe ratio is about 0.6, so only strategies with Sharpe $\gtrsim 1.2$ can be confirmed at 5%. Event and cross-sectional designs gain power from breadth, but a 15-name universe limits that gain.

2 Data and execution realism

The supplied folder has 21 datasets (88.5 GB, 4.74 billion rows; the audit is in `reports/dataset_audit/ASSESSMENT.md`). Table 1 lists the datasets used and their roles.

Table 1: Datasets and their role in the backtests.

Dataset	Content	Role here
EOD option chain	bid/ask, IV, Greeks, OI for 20 underlyings	Daily signals; T3 fills; daily marks
Option trades + NBBO	every sampled print with the prevailing bid/ask and sizes	T2 fills, effective spreads, signed order flow
Equity NBBO	quotes 09:29–09:35 and 15:55–16:01 for 17 names (no GOOGL)	Stock hedges and entries near the open and close
Equity auction-window trades	prints around the opening and closing crosses	T1 auction fills
Context	daily equity OHLC, factors, rates, VIX, earnings dates, fundamentals	Signals, corporate actions, benchmarks

Table 2: Fill tiers T0–T3.

Tier	Mechanism	Used for
T0	Cash settlement on the SPX close (SPXW PM only)	Option exits held to expiry
T1	Official opening or closing cross price	Stock entries and exits
T2	First observed NBBO strictly after decision time + latency, within 10 s; buy at the ask, sell at the bid; size $\leq$ displayed	Intraday option and stock orders
T3	EOD bid/ask (buy ask, sell bid)	Daily option strategies

2.1 Fill tiers

Every backtest declares how each order is filled (Table 2). Passive fills are never assumed. For multi-leg orders, every leg must fill under T2 or the trade is skipped, and skips are counted.

In the T2 rule, the decision time is τ and the latency is L. The wave-2 pre-registration sets $L = 2$ s as primary, with 0.5 s and 10 s as sensitivities. A T1 auction fill also requires participation of at most 1% of the trailing-20-session median print size.

2.2 Measured data facts

The following facts were measured before the designs were written, and they shaped them.

- **Clock.** The option tape is in US/Eastern time. SPY option prints start at 09:30:00.
- **Capped sample.** The tape holds about 500,000 prints per day for SPY, QQQ and TSLA, and SPY options are missing on 193 of 625 development days. In a 10-second window only 30–45 SPY contracts and 3–30 contracts per single name show a print, so quotes are visible only at sampled prints. Flow signals therefore use ratios, never levels.
- **Effective versus quoted spreads.** SPXW prints well inside the quoted spread (effective/quoted half-spread ratio 0.57 at 0DTE ATM and 0.34 at 8–30 DTE). SPY and QQQ print near the touch (0.75–0.93); see Figure 2.
- **Auction data.** Closing-cross prints equal the official daily close on 100% of tested days. XOM and LLY usually lack a primary-exchange cross, and GOOGL has no equity quote or auction data.
- **Share-class mixing.** The field `sym_root` mixes share classes. JPM preferred shares near $20

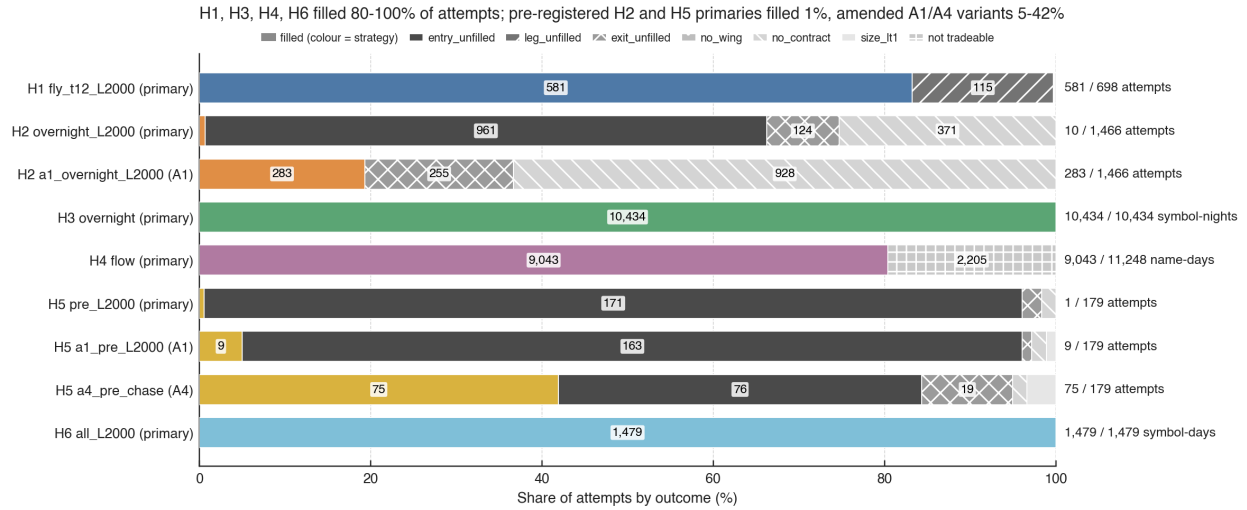


Figure 1: Share of intended trades that fill, by wave-2 design (`wave2/figures/fig01_fill_funnel.png`). Fill availability, not the signal, limits several designs.

Table 3: Wave-3 transaction-cost schedule.

Item	Cost
Option commission	\$0.65 per contract per side
SPX/SPXW exchange + OCC + ORF	\$0.66 + \$0.025 + \$0.003 per contract
Other option taker fees	\$0.50 + \$0.028 per contract
Stock	\$0.0035 commission + \$0.0002 clearing + \$0.0010 auction fee per share
Sells	SEC fee \$27.80 per \$1M, FINRA TAF \$0.000166 per share (capped at \$8.30 per trade)
Borrow	0.40% per year on short notional

appear in both the JPM quote stream and the auction prints, and both are filtered.

- **EOD snapshot time.** EOD chain quotes match the tape's last NBBO before 16:00 or 16:15 only 7–21% of the time, so the snapshot time cannot be determined.

2.3 Transaction costs

Wave 2 used placeholder option fees. Wave 3 uses a realistic schedule (Table 3), which should be verified against a broker statement. Fees follow a public schedule, not executed broker statements.

3 Research protocol

3.1 Waves

Table 4 summarizes the four waves. Out-of-sample data were withheld from the engines until a freeze commit in git.

For waves 2 and 3 the development and test periods are as follows. Wave 2 uses development 2022-01-03 to 2024-06-28 and a holdout from 2024-07-01 to 2024-12-31. Each primary specification

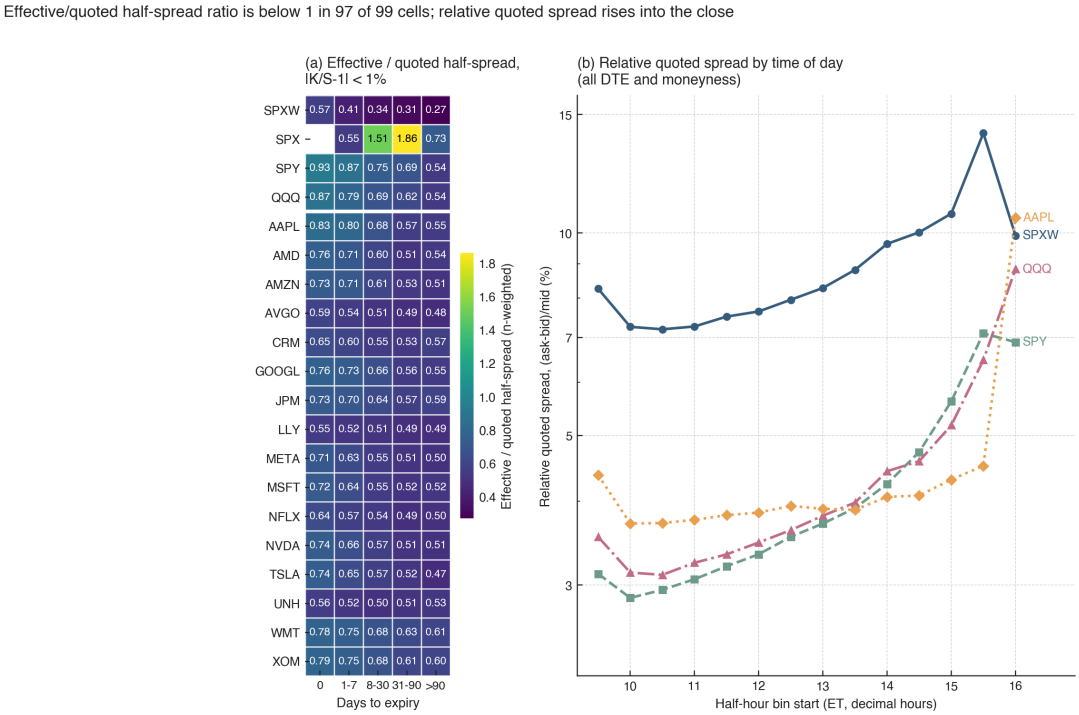


Figure 2: Execution-cost atlas: effective against quoted spreads by underlying, days to expiry and time of day (`wave2/figures/fig02_cost_atlas.png`).

Table 4: Research waves and out-of-sample designs.

Wave	Design	Out-of-sample
0	72 EOD configurations (verticals, condors) on SPY/QQQ/SPX, fixed 60/20/20 split	2024-06 to 2024-12 test
1	220 EOD configurations, rolling 252/63-session walk-forward	6 folds, 375 sessions
2	6 intraday-fillable hypotheses (H1–H6) and 2 foundations (F1, F2)	2024-07 to 2024-12; tapes not downloaded until the freeze commit `d35c50e`
3	10 strategies (S1–S10), realistic fees	2024 test year; engines refused dates after 2023-12-31 until the freeze commit `e127299`

runs once on the holdout, with no changes. In wave 3 the confirmatory strategies (S1, S2, S5, S6, S8, S9, S10) use development 2022-01-03 to 2023-12-29 and a test year of 2024. The exploratory strategies (S3, S4, S7) were derived from wave-2 results (H4 and H1), run on the full 2022–2024 sample, and cannot be confirmed here.

3.2 Pre-registration and amendments

Each strategy was written down before its engine ran, in the order economic intuition, signal, portfolio construction, execution (fill tier), risk, evaluation (`wave2/prereg/PREREGISTRATION.md`, `wave3/PREREGISTRATION.md`). Each strategy has one primary specification. Secondary variants are reported and corrected for multiplicity.

Changes made after a run are dated in the pre-registration files with the reason.

- Amendments A1–A4 (wave 2) respond only to measured fill availability. For example, chosen-strike straddles filled on about 1% of nights.
- Amendment B1 (wave 3) changes the denominator of one per-trade test (S5). It was made after seeing development totals and is disclosed as such.
- The original specifications are always reported alongside the amended ones.

The wave-3 test year is not a virgin holdout. Wave-2 results on 2024 had been seen, though none of the wave-3 signals had been computed on that year. S8 is closest to wave-2 H3 and is marked "partially informed".

3.3 Inference

- Newey–West t-statistics with 5 lags (Newey and West, 1987) and stationary block-bootstrap confidence intervals with mean block length 10 (Politis and Romano, 1994), using 5,000 draws.
- Event returns are clustered by date. Cross-sections use Fama–MacBeth regressions (Fama and MacBeth, 1973).
- Romano–Wolf stepdown p-values (Romano and Wolf, 2005) are computed within each strategy family.
- Deflated Sharpe ratios (Bailey and López de Prado, 2014) are computed against all 88 wave-2 and wave-3 variants. Waves 0 and 1 add 292 earlier trials, so the true multiplicity is higher still (see also Harvey et al., 2016).

3.4 Kill rule

A primary is rejected if any of the following holds:

1. its development mean is ≤ 0;
2. its out-of-sample mean is ≤ 0;
3. its development result disappears once the five best days are removed.

Surviving the kill rule is necessary, not sufficient. A result would be called "confirmed" only with a deflated Sharpe ratio near 1 and a Romano–Wolf $p < 0.05$. Nothing here meets that.

3.5 What 2022–2024 can detect

With three years of data the standard error of an annualized Sharpe ratio is about 0.6. Only strategies with Sharpe $\gtrsim 1.2$ can therefore be confirmed at 5%. Event and cross-sectional designs

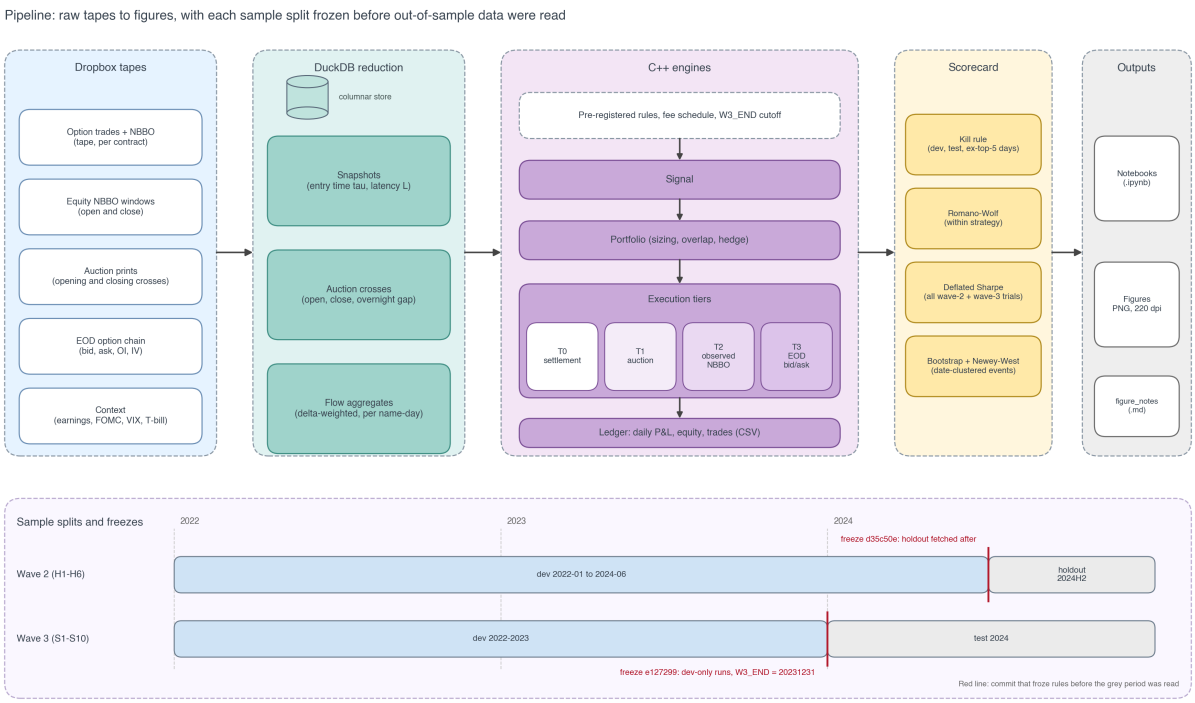


Figure 3: Research pipeline from vendor tapes to engines, inference and figures (`wave3/figures/fig00_pipeline.png`).

gain power from breadth, and the 15-name universe limits that gain.

4 Implementation

Engines are C++17/20 (`wave2/cpp`, `wave3/cpp`). Inference is in Python (`wave2/analysis.py`). Figures and tables live in executed notebooks (`wave2/notebooks`, `wave3/notebooks`). Table 5 gives the code layout.

4.1 Snapshots, forward and strike selection

For a decision time τ and latency L, `extract.py` stores two quotes per contract c:

$$q_c^{\text{info}} = \text{last quote in } [\tau - 120\,\text{s},\ \tau), \tag{1}$$

$$q_c^{\text{fill}} = \text{first quote in } [\tau + L,\ \tau + L + 10\,\text{s}]. \tag{2}$$

The first is information known at τ. The second is the T2 fill quote. Latency is run at $L \in \{0.5\,\text{s}, 2\,\text{s}, 10\,\text{s}\}$, and a 120 s "chase" window is stored for the amended exits.

Strikes use the put–call-parity forward, so no spot feed is needed for SPX:

$$F = K^* + C - P, \qquad K^* = \arg\min_K |C_K - P_K|, \tag{3}$$

where C and P are call and put mids at the strike K^*, taken from the pre-decision quotes in (1).

Table 5: Code layout.

Path	Role
`wave2/fetch.sh`, `fetch_dev.sh`, `fetch_holdout.sh`	Manifest-driven Dropbox download with byte-count verification
`wave2/extract.py`	DuckDB reduction: decision-time quote snapshots, auction crosses, flow, cost atlas
`wave2/context.py`	Small daily inputs (sessions, equity, earnings, H4 signals, GEX, F2 chain)
`wave2/cpp/common.hpp`	CSV loader, sessions, T2 snapshot lookup, parity forward, Black–Scholes, ledger
`wave2/cpp/equity.hpp`	Auction crosses (validated against official open/close), corporate actions
`wave2/cpp/{f2,h1..h6}_*.cpp`	One engine per strategy: signal, portfolio, execution, accounting
`wave2/analysis.py`, `scorecard.py`	Inference and pre-registered decisions
`wave2/notebooks/wave2_report.ipynb`, `figures/`	Report notebook and figures
`wave3/context3.py`	SPXW and single-name chains, OI, flow measures, VIX
`wave3/cpp/w3.hpp`	Realistic fees, `W3_END` guard, beta, mark-to-market stock book
`wave3/cpp/{s_events,s3_flow,s5_spxw_vix, s6_straddle_gs,s7_pin,s_calendar}.cpp`	Wave-3 engines
`wave3/scorecard3.py`, `notebooks/wave3_report.ipynb`, `figures/`	Scorecard, notebook, figures
`src/options_lab/`, `wave01/`	Waves 0 and 1 (Python EOD engine, 64 tests)

4.2 Accounting

Each strategy runs a $1M reference account per variant, and idle cash earns the 1-month T-bill. Stock positions (wave 3) are marked to the official close each day. Option positions are marked to the EOD mid and realized at settlement or exercise. Reported returns are daily excess returns.

4.3 Checks

- **Unit self-checks.** The payoff identities, implied-volatility round-trips and statistical routines each carry an assert.
- **Accounting reconciliations.** The 0DTE settlement close matches the intraday parity forward to within about 2 bp on average. The wave-3 stock book reconciles exactly: price P&L $111,375 − fees $861 + dividends $797 = ledger $111,311. The H2 P&L decomposition identities hold to within $0.005. A hand-recomputed S5 trade matches the engine to the cent.
- **External benchmark.** The put-write engine tracks the published Cboe PUT index with monthly correlation 0.972 and tracking error 2.3% per year (F2).
- **Bugs caught by mechanics checks** before any out-of-sample run: preferred-share prices contaminating the JPM stream; a session-index leak in the wave-3 date guard; and a missing equity-quote window that blanked H6.

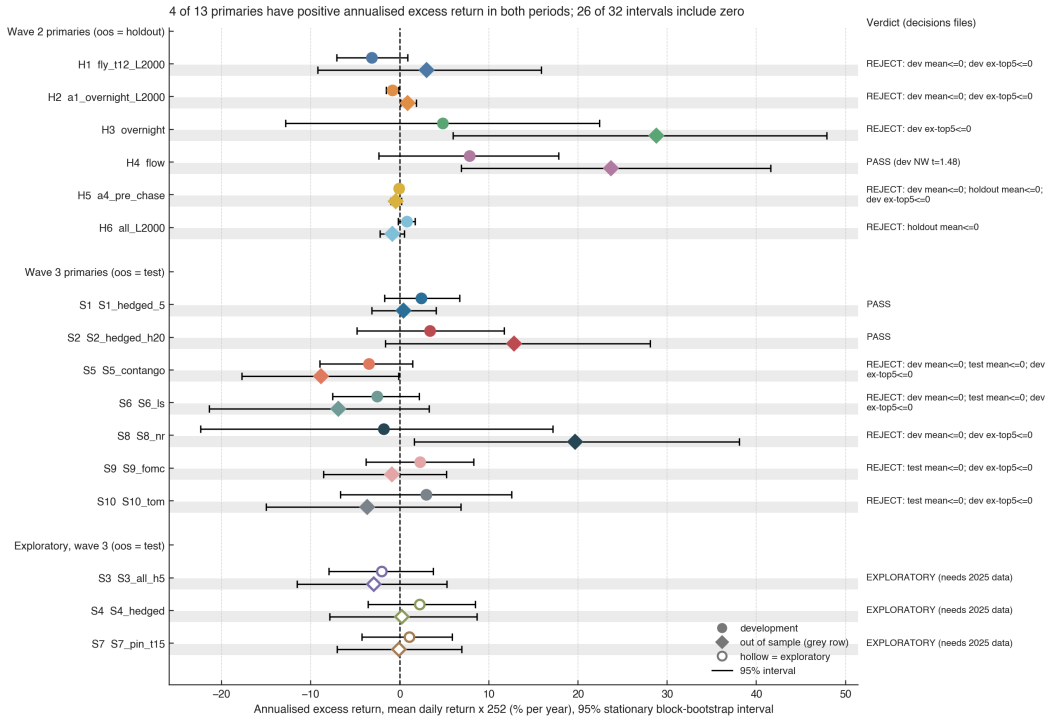


Figure 4: Forest plot of all primary specifications (`wave3/figures/fig11_forest_all.png`).

5 Results overview

Table 6 lists each strategy with its primary (or amended) specification. "Dev" is the development period and "OOS" is the wave-2 2024H2 holdout or the wave-3 2024 test. Returns are annualized excess returns on the $1M reference account. H4 survives the kill rule but is not significant, and its wave-3 re-implementation S3 is consistent with it. S1 and S2 also survive the kill rule without being significant. All other primaries are rejected.

Table 6: Primary (or amended) specification of each strategy. Dev is the development period; OOS is the wave-2 2024H2 holdout or the wave-3 2024 test. Returns are annualized excess returns on the $1M reference account. Source: README Section 4.

Wave	Strategy	Dev ann. / Sharpe / NW t	OOS ann. / Sharpe	Decision
0	EOD put/call credit spreads, condor, debit vertical (4 structures)	selected on dev	−0.4% to −4.5%	Reject all
1	Rolling EOD grid (5 structures)	selected per fold	−1.0% to −3.7%; call credit spreads t = −2.21	Reject all
2	F2 put-write engine check	corr 0.972 with Cboe PUT	n/a	Engine valid
2	H1 0DTE SPXW iron fly, held to settlement	−3.1% / −0.79 / −1.42	+3.0% / 0.75	Reject

Wave	Strategy	Dev ann. / Sharpe / NW t	OOS ann. / Sharpe	Decision
2	H2 overnight delta-hedged short straddle (A1)	−0.8% / −1.39 / −2.22	+0.9% / 2.44	Reject; original rule untestable (10 fills)
2	H3 overnight equity drift (auctions)	+4.8% / 0.29 / 0.50	+28.8% / 2.44	Reject (top-5 days)
2	**H4 option order flow → next-day returns**	**+7.8% / 0.90 / 1.48**	**+23.7% / 3.26**	**Survives kill rule; not significant**
2	H5 pre-earnings straddle (A4)	−0.1% / −0.18 / −0.30	−0.5% / −2.06	Reject; original rule untestable (1 fill)
2	H6 end-of-day momentum, SPY/QQQ	+0.8% / 0.91 / 1.50	−0.9% / −1.37	Reject
3	S1 pre-earnings premium, beta-hedged	+2.4% / 0.88 / 1.08	+0.4% / 0.17	Survives kill rule; not significant
3	S2 earnings-announcement-return drift	+3.4% / 0.64 / 0.87	+12.8% / 1.63	Survives kill rule; not significant
3	S3 option flow, realistic fees (1-day hold, exploratory)	+8.6% / 0.96 / 1.38	+12.9% / 1.71	Exploratory; consistent with H4
3	S4 pre-announcement option flow (exploratory)	+2.2% / 0.45 / 0.66	+0.2% / 0.05	Exploratory; no effect
3	S5 SPXW iron fly held to settlement, VIX-timed	−3.5% / −1.08 / −1.37	−8.9% / −1.59	Reject
3	S6 RV − IV straddles exited by exercise	−2.6% / −0.73 / −1.06	−6.9% / −0.95	Reject
3	S7 0DTE pin at max-OI strike (exploratory)	+1.0% / 0.28 / 0.38	−0.1% / −0.03	Exploratory; no pin effect
3	S8 daytime-reversal frequency	−1.8% / −0.13 / −0.20	+19.6% / 1.69	Reject
3	S9 FOMC pre-announcement (24 events)	+2.3% / 0.53 / 0.74	−0.9% / −0.25	Reject
3	S10 turn-of-the-month	+3.0% / 0.36 / 0.55	−3.6% / −0.61	Reject

Equity curves are in Figure 5 and Figure 6. Full scorecards are `wave2/results/decisions.csv` and `wave3/results/decisions3.csv`.

6 Strategy studies

Each study follows the same structure: economic intuition and literature, specification (signal and trading rules), implementation, and out-of-sample testing. Numbers are net of the stated costs.

6.1 Waves 0 and 1: EOD defined-risk option spreads

Economic intuition and literature. Index options carry a variance risk premium (Carr and Wu, 2009; Bollerslev et al., 2009). Defined-risk short-premium structures might harvest it, and debit verticals might express trend. These two waves tested that idea on end-of-day data, before the intraday tape was used.

Specification.

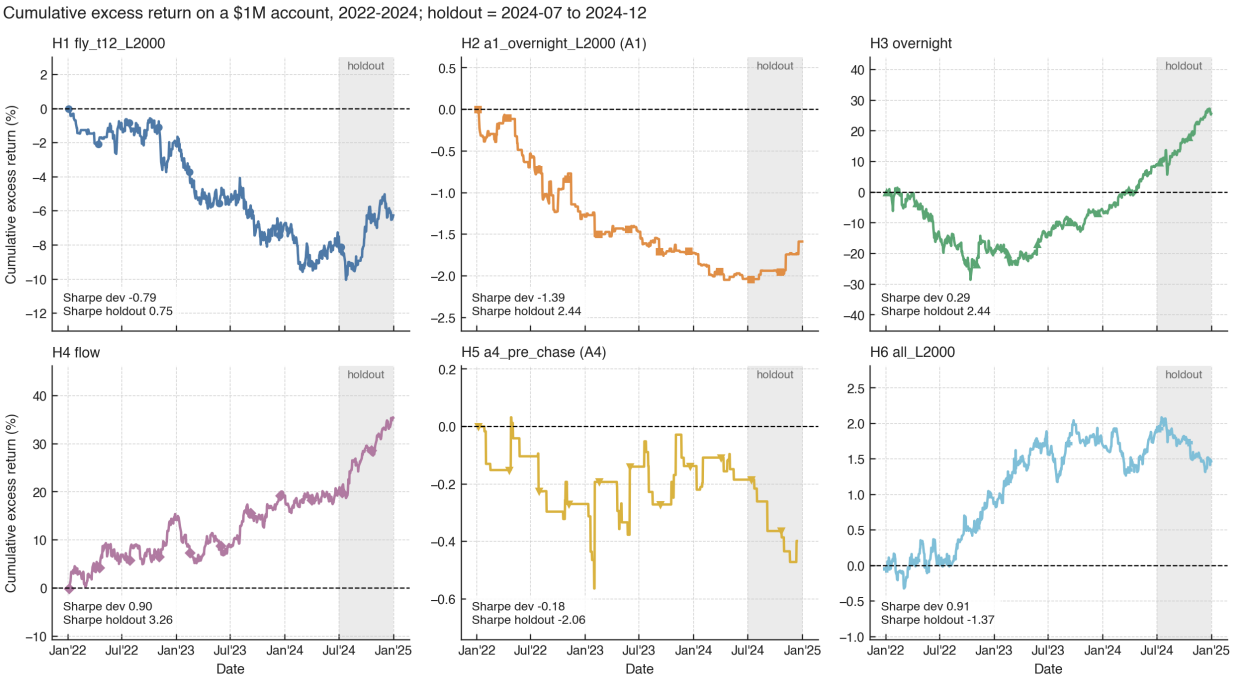


Figure 5: Wave-2 equity curves (`wave2/figures/fig04_equity_curves.png`).

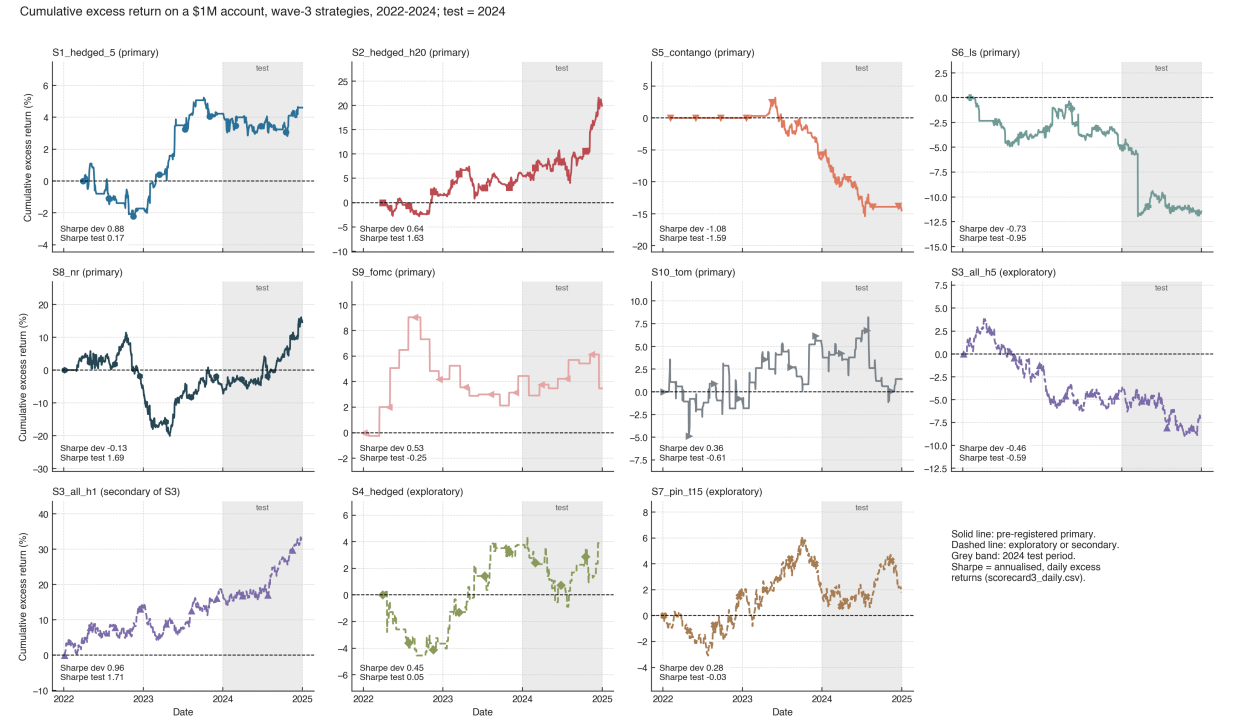


Figure 6: Wave-3 equity curves (`wave3/figures/fig12_w3_equity.png`).

Table 7: Waves 0 and 1: out-of-sample annualized excess returns.

Wave	Design	Out-of-sample	Decision
0	4 structures, selected on dev	-0.4% to -4.5%	Reject all
1	5 structures, selected per fold	-1.0% to -3.7%; call credit spreads $t = -2.21$	Reject all

- *Structures:* put and call credit spreads, iron condors, and call and put debit verticals on SPY, QQQ and SPX.
- *Signal and filters:* short-leg delta 0.15–0.30 and 30–45 DTE. Filters include IV percentile, implied-minus-realized volatility, trend and skew.
- *Exits:* at 50% of the credit, a 2× loss stop, or at 5 DTE or less.
- *Execution tier:* T3, next EOD snapshot at buy-ask and sell-bid, with per-leg fees. Signals use the prior close.
- *Risk limits:* defined by construction (spread width).

Implementation. The Python EOD engine is in `src/options_lab` (64 tests). Wave 0 covers 72 configurations with a fixed 60/20/20 split and a 2024-06 to 2024-12 test. Wave 1 covers 220 configurations with a rolling 252/63-session walk-forward (6 folds, 375 sessions). Details are in `reports/06_final_research_report.md` and `reports/rolling_research/`.

Out-of-sample testing and results. Table 7 gives the out-of-sample results. All four wave-0 structures were rejected on the fixed test window, and the deflated Sharpe ratio is about 0 over 72 trials. In wave 1, rolling selection over 220 configurations gave negative excess returns for every structure across the 6 folds. Call credit spreads were the worst, with Newey–West $t = -2.21$. Entering and exiting across the full spread cost more than any premium these structures captured.

6.2 F1 and F2: execution-cost atlas and engine validation

Economic intuition and literature. Quoted spreads can overstate the costs that liquidity takers pay (Muravyev and Pearson, 2020). Wave 2 therefore began with a measurement study (F1) and an engine check (F2). Neither is an alpha test. If the engine cannot reproduce a published, mechanically defined index, its strategy P&L is not credible.

Specification.

- **F1 (measurement).** For each trade with a valid NBBO, compute the effective half-spread ratio and the relative quoted spread,

$$\rho = \frac{|p - m|}{(a - b)/2}, \qquad s_{\text{rel}} = \frac{a - b}{m}, \tag{4}$$

  where p is the trade price, m the NBBO mid, a the ask and b the bid. Results are bucketed by root, DTE $\{0, 1\text{–}7, 8\text{–}30, 31\text{–}90, > 90\}$, moneyness $|K/S - 1| \in \{< 1\%, 1\text{–}3\%, 3\text{–}7\%, > 7\%\}$ and half-hour, with S the prior close. A timing check compares the EOD-chain bid/offer with the tape's last NBBO before 16:00 and 16:15 for SPY, SPXW and AAPL.
- **F2 (replication).** On each third-Friday expiry, sell one SPXW PM put expiring next month. The strike is the listed strike nearest to and not above the SPX close. The fill is the EOD bid

Table 8: F2: put-write engine against the Cboe PUT index.

Statistic	Value
Monthly return correlation	0.972
Tracking error	2.3% per year
Mean monthly gap	−9 bp
Monthly rolls	34

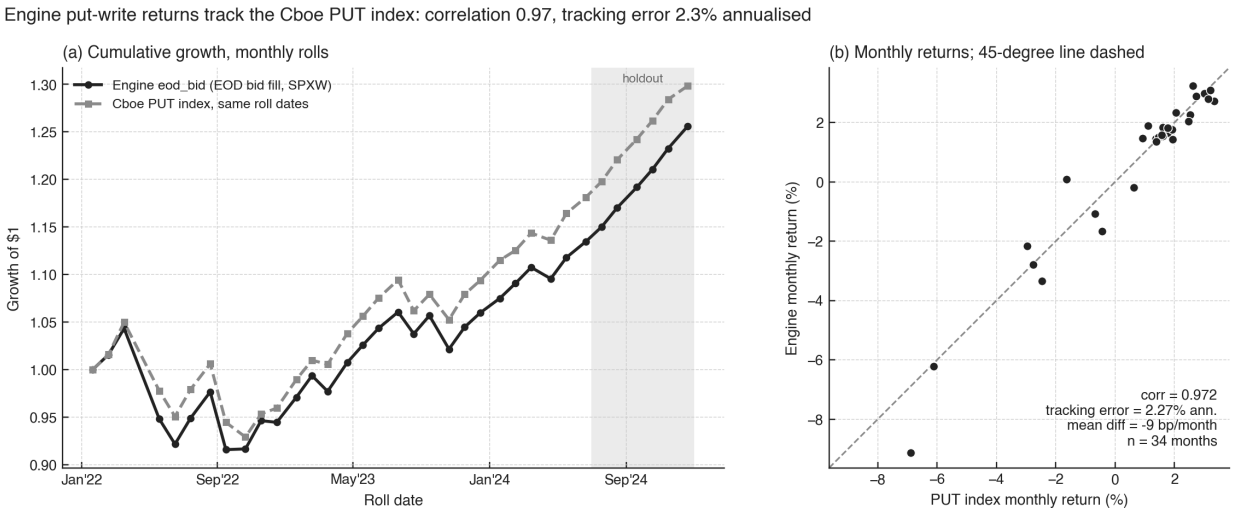


Figure 7: F2: put-write engine against the Cboe PUT index (`wave2/figures/fig03_putwrite_validation.png`).

(T3; the mid is a sensitivity). Collateral is the full strike notional in T-bills. Hold to expiry and settle on the SPX close (T0).

- **Execution tier:** T3 entry and T0 exit for F2. **Risk:** the put is fully cash-secured.

Implementation. F1 uses `wave2/extract.py` to build the atlas from about 1.5 billion sampled development-period prints. F2 is run by the common engine (`wave2/cpp/common.hpp` and the F2 engine). The F2 tape sensitivity uses the first bid within 120 s of the 15:55 sale (amendment A3). The primary EOD variant is unchanged. Differences from the Cboe PUT index are expected because PUT uses AM settlement and a VWAP sale window. These differences are documented rather than fitted.

Out-of-sample testing and results. Neither study has a holdout test. For F1, SPXW prints well inside the spread, SPY and QQQ print near the touch, and single names fall in between (Figure 2). The effective/quoted half-spread ratio is 0.57 at 0DTE ATM and 0.34 at 8–30 DTE for SPXW, and 0.75–0.93 for SPY and QQQ. Other traders' effective spreads are not our fills, so T2 still crosses the full spread.

For F2, the monthly cash-secured SPXW put-write tracks the Cboe PUT index (Table 8). The mean gap is in line with PUT's AM settlement and VWAP sale window, which we do not replicate. We read this as evidence that the option accounting is sound.

Table 9: H1: iron fly at 12:00 held to settlement.

Sample	Ann. excess	Sharpe	Other
Development	−3.1%	−0.79	NW $t = -1.42$; trade mean −4.0%
Holdout (2024H2)	+3.0%	0.75	CI −9% to +16%

6.3 H1: 0DTE SPXW premium held to settlement

Economic intuition and literature. Same-day index options may embed a variance premium, because demand for same-day convexity from retail traders and hedgers can exceed the risk that dealers are compensated for (Bandi et al., 2023; Beckmeyer et al., 2023; Vilkov, 2023). Holding to cash settlement pays the spread once and collects the premium without a second spread.

Specification. The primary signal is unconditional: every day with a 0DTE SPXW expiry and valid quotes. The forward F_τ is the parity forward of (3), built from each contract's last quote in $[\tau - 120\,\text{s}, \tau)$.

- *Portfolio:* short iron fly. Sell the call and put at K^* and buy the wings at $K^* \pm w$, where w is the listed strike nearest 1.5× the straddle mid (minimum 5 points). One trade per day.
- *Sizing and risk:* contracts n are sized so that the maximum loss satisfies

$$(w - \text{credit}) \times 100 \times n \;\le\; 0.005\,E, \tag{5}$$

  where E is account equity. Risk is defined by construction. We report the worst day, ES97.5 and the share of P&L from the best 5 days.
- *Primary τ:* 12:00 ET. Secondaries are 10:00, 14:00 and 15:00, and a short iron condor (shorts at $K^* \pm 1\times$ the straddle, wings at $K^* \pm 2\times$ the straddle).
- *Secondary signal:* a filter that trades only when the implied move (ATM straddle mid divided by F) exceeds the trailing-20-session median realized move.
- *Execution tier:* T2 entry (sell at the bid, buy at the ask) and T0 exit at `sp500_cash_level`.

Implementation. The engine is `wave2/cpp/h1_0dte.cpp`. Fills are T2 on all four legs, and all legs must fill. Four-leg fills succeeded on about 80% of days, so H1 is the one wave-2 option design with adequate fill availability and no amendment. The 0DTE settlement close matches the intraday parity forward to within about 2 bp on average.

Out-of-sample testing and results. The development mean return on risk is −4.0% per trade (date-clustered $t = -1.22$), with annual excess return −3.1% and Sharpe −0.79 (Table 9). The holdout is +3.0% (confidence interval −9% to +16%). Later entries are closer to zero, and the condor is about 0. None of the 15 variants is significant after Romano–Wolf correction (Figure 8). The decision is reject.

6.4 H2: overnight versus intraday option premium

Economic intuition and literature. Muravyev and Ni (2020) find that delta-hedged option returns are strongly negative overnight. Jones and Shemesh (2018) report lower option returns over non-trading periods. Option prices seem to spread expected variance evenly over calendar time, while most variance arrives during the session, so short sellers of overnight variance should be paid.

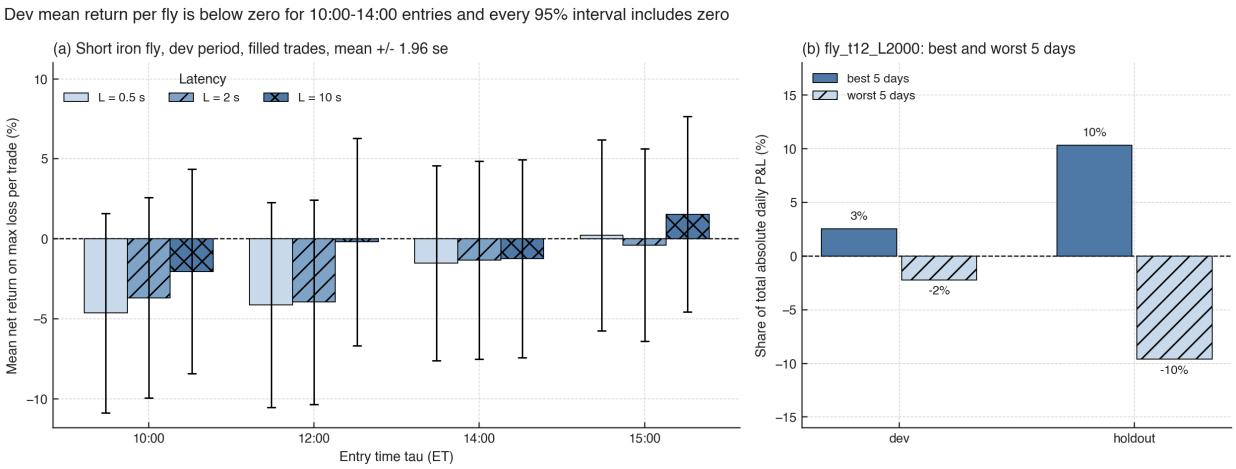


Figure 8: H1: return on risk by entry time and latency (`wave2/figures/fig05_h1_entry_latency.png`).

Specification.

- *Signal:* unconditional, every session pair (close of t to open of $t+1$). A secondary restricts to weekends.
- *Portfolio:* short one ATM SPY or QQQ straddle (strike nearest F, first expiry with DTE ≥ 7), delta-hedged with shares at entry using the Black–Scholes delta from IV inverted at the mid. The size n limits the straddle's gamma loss on a 3% gap to 0.5% of equity,

$$n = \left\lfloor \frac{0.005\,E}{\frac{1}{2}\,\Gamma\,(0.03\,S)^2 \times 100} \right\rfloor, \tag{6}$$

  computed for SPY and QQQ independently.
- *Execution tier:* option entry at 15:58:00 and exit at 09:33:00 the next session (both T2). The stock hedge is T2 against the equity NBBO at the same times.
- *Risk limits:* gap risk is bounded by the sizing rule, not by construction. We report the worst nights.
- *Contrast:* the same trade held intraday, 09:33 to 15:58.

Implementation. The engine is `wave2/cpp/h2_overnight.cpp`. Options are T2, and the stock hedge is T2 against the equity NBBO, which exists exactly in the open and close windows. The pre-registered contract rule filled 10 nights, so the original primary is untestable. Amendment A1 (made only because of measured fill availability) chooses among contracts with observed quotes and allows a 120 s exit chase. The P&L decomposition identities hold to within \$0.005.

Out-of-sample testing and results. The A1 overnight trade lost 1.25% of premium per night in development (date-clustered $t = -4.0$). Table 10 decomposes the result (Figure 9). In this sample the option-value decline appears mostly during the session, which is the opposite of Muravyev and Ni (2020). The 09:33 opening quote may carry an opening-IV premium. The spread alone is about four times the overnight premium. Annual excess return is -0.8% (Sharpe -1.39, NW $t = -2.22$) in development and $+0.9\%$ (Sharpe 2.44) in the holdout. The decision is reject.

Table 10: H2 (A1): components of P&L, % of premium per night, development.

Component	Overnight	Intraday
Mid-to-mid option P&L	+0.18	+1.58
Spread cost	−0.76	−0.77
Hedge	−0.33	−0.33
Fees	−0.33	−0.34

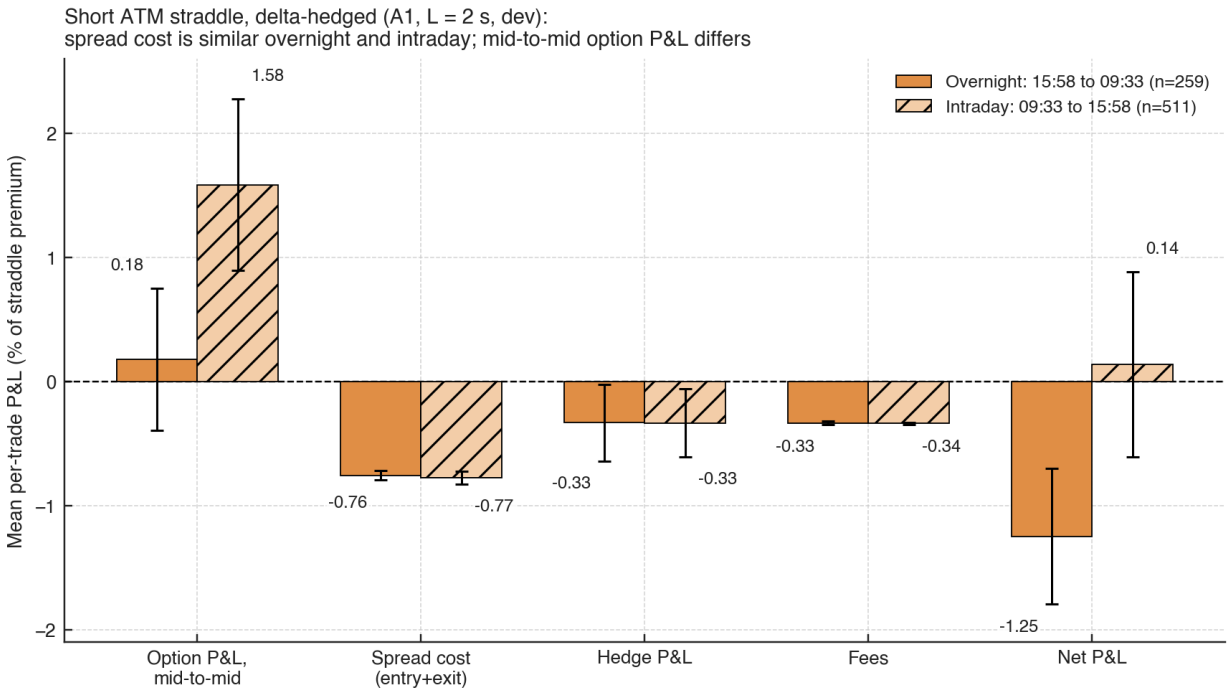


Figure 9: H2: overnight and intraday P&L decomposition (`wave2/figures/fig06_h2_decomposition.png`).

Table 11: H3: overnight equity drift.

Sample	Ann. excess	Sharpe	NW t
Development, overnight	+4.8%	0.29	0.50
Development, intraday leg	+15.3%		
Holdout (2024H2), overnight	+28.8%	2.44	

6.5 H3: overnight equity drift with auction fills

Economic intuition and literature. A large share of the equity premium may accrue overnight, through a tug of war between overnight and intraday clienteles (Lou et al., 2019; Cliff et al., 2008). Auctions allow entry and exit at a single price without crossing a spread.

Specification.

- *Signal:* the primary is an always-on long overnight position. The secondary is overnight momentum: rank names on their trailing-21-session mean overnight return and hold only the top third.
- *Portfolio:* equal weight across 14 single names plus SPY and QQQ, 100% gross, rebuilt daily. META and GOOGL are excluded because they lack auction coverage.
- *Execution tier:* T1. Buy at the closing cross and sell at the next opening cross, with fees on each side and participation at most 1% of lagged cross size. A sensitivity crosses the spread at 15:59 and 09:31 instead of using auctions.
- *Contrast:* the intraday (open to close) leg.

Implementation. The engine is `wave2/cpp/h3_overnight_eq.cpp`. Fills are T1. Splits are handled with `share_adj_factor`, and ex-dividend nights are credited the dividend from the gap between total and price return. Auction crosses are validated against the official open and close.

Out-of-sample testing and results. Development overnight returns are +4.8% per year (CI −12.8% to +22.2%; Sharpe 0.29, NW $t = 0.50$), against +15.3% for the intraday leg, the reverse of the documented pattern (Figure 10). The 2024H2 result (+28.8%, Sharpe 2.44) coincides with a broad rally and does not survive removing the five best days. The decision is reject.

6.6 H4 and S3: option order flow and next-day stock returns

Economic intuition and literature. Informed traders use options for leverage and concealment, so option order flow and implied-volatility asymmetries can lead stock prices (Easley et al., 1998; Pan and Poteshman, 2006; Cremers and Weinbaum, 2010; Xing et al., 2010; Johnson and So, 2012). The signal is built from the sampled tape using ratios only, which keeps it robust to the sampling cap.

Specification.

- *Primary signal:* the delta-weighted order imbalance of day t,

$$\mathrm{OI}^{\mathrm{flow}}_{i,t} = \frac{\sum_k \delta_k \,\mathrm{size}_k \,\mathrm{side}_k}{\sum_k |\delta_k| \,\mathrm{size}_k}, \tag{7}$$

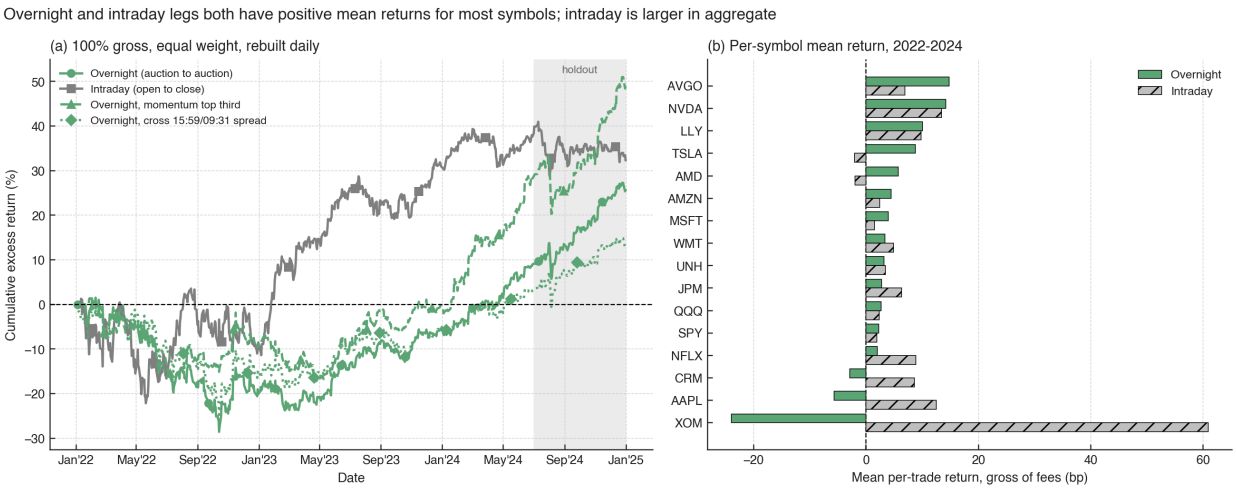


Figure 10: H3: overnight and intraday returns (`wave2/figures/fig07_h3_overnight_intraday.png`).

summed over regular-hours option trades k on name i. The side is $+1$ if the price is above the mid and -1 if it is below (the quote rule), and δ_k is the same-day EOD delta.

- *Secondary signals:* the IV spread (OI-weighted mean of $\mathrm{IV}_{\mathrm{call}} - \mathrm{IV}_{\mathrm{put}}$ over matched strike and expiry pairs, 10–60 DTE); the smirk (OTM-put IV at K/S nearest 0.95 within $[0.80, 0.95]$ minus ATM-call IV nearest $K/S = 1$, 10–60 DTE, with a negative sign expected); and an equal-weight z-composite.
- *Portfolio:* 15 single names (16 minus META), cross-sectional demeaned ranks scaled to 100% gross, dollar neutral, held from the $t+1$ opening cross to the $t+2$ opening cross.
- *Execution tier:* T1 opening-auction prints, with fees and short borrow. GOOGL has no auction data, so it is ranked but traded at the 09:31 window NBBO.
- *Evaluation:* the Fama–MacBeth slope of the next open-to-open return on the cross-sectionally z-scored signal (Newey–West t) and the long-short Sharpe ratio.
- *S3 variants (wave 3, exploratory):* `all` (equal to H4); `open`, flow on contracts whose OI rises (the timing of OI is an assumption); and `short_otm`, with DTE of at most 30 and $|\delta| < 0.4$. Holding periods are 1 day and 5 days (5 overlapping sub-books). S3 has no primary.

Implementation. The wave-2 engine is `wave2/cpp/h4_info.cpp`. It was re-implemented in wave 3 with realistic fees as `wave3/cpp/s3_flow.cpp`. Signals come from `wave2/context.py` and `wave3/context3.py`. S3 was derived from a wave-2 result, so it runs on the full 2022–2024 sample and cannot be confirmed here.

Out-of-sample testing and results. Table 12 reports the results. The Fama–MacBeth slope on the 1σ flow signal is 3.6 bp per day in development ($t = 1.63$) and 12 bp in the holdout ($t = 2.32$; Figure 11). The effect is gone within a week: 5-day holds are negative for every flow definition (Figure 12). The opening-flow and short-dated-OTM refinements do not improve on raw flow in development.

We classify this as a lead. It survives the kill rule and realistic fees. Its development evidence, however, is not distinguishable from the best of the trials we ran, and it is the one wave-2 result we went on to refine. The sample is 15 names, typically about 10 tradeable.

Table 12: H4 and S3: long-short option-flow book.

Sample	Ann. excess	Sharpe	NW t	Other
Wave 2 dev	+7.8%	0.90	1.48	DSR 0.17; Romano–Wolf $p = 0.36$
Wave 2 holdout (2024H2)	+23.7%	3.26		bootstrap CI +7% to +42%
Wave 3 dev, realistic fees, 1-day hold	+8.6%	0.96	1.38	
Wave 3 test (2024), 1-day hold	+12.9%	1.71	1.82	

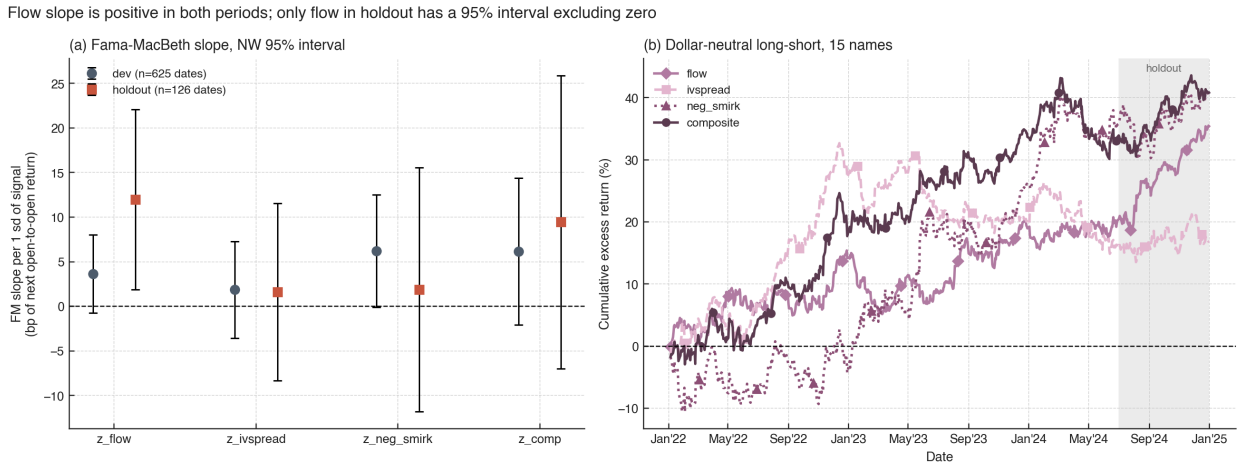


Figure 11: H4: Fama–MacBeth slopes by signal and period (`wave2/figures/fig08_h4_fama_macbeth.png`).

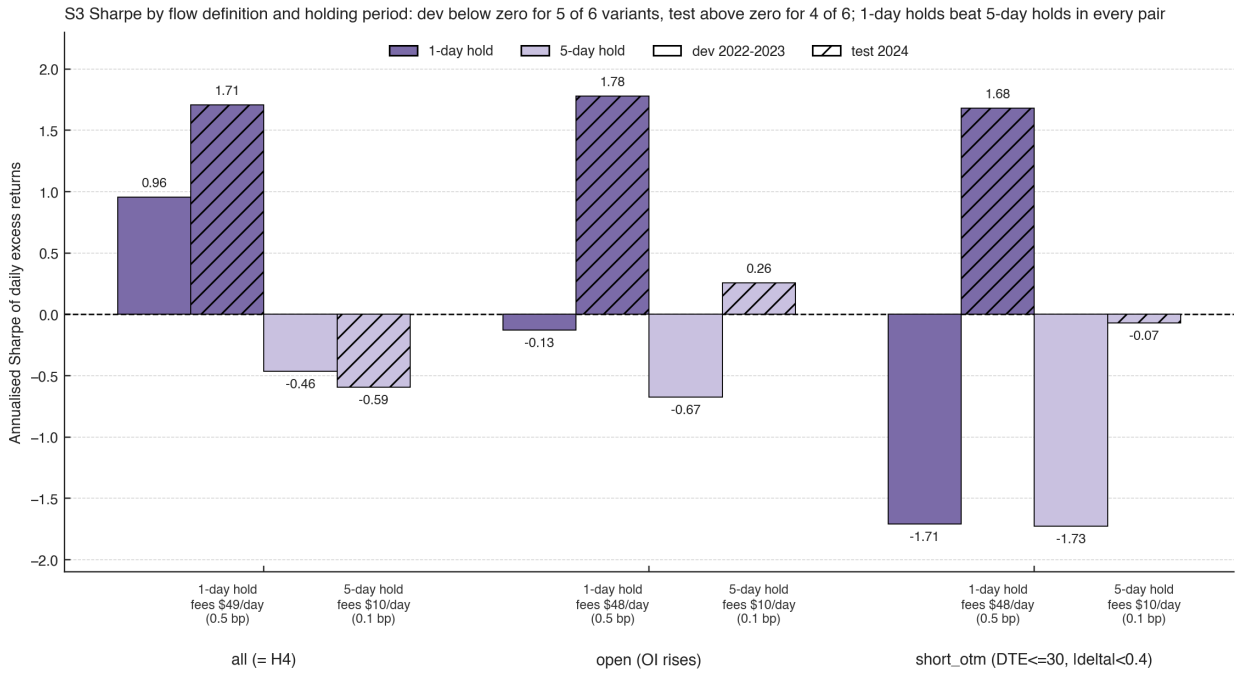


Figure 12: S3: decay of the flow effect with holding period (`wave3/figures/fig15_s3_decay.png`).

Table 13: H5 (A4): pre-earnings straddle.

Sample	Ann. excess	Sharpe	NW t
Development	−0.1%	−0.18	−0.30
Holdout (2024H2)	−0.5%	−2.06	

6.7 H5: pre-earnings implied-volatility run-up

Economic intuition and literature. Uncertainty about an announcement is priced gradually, so implied volatility rises into the event. Straddles bought days before and sold before the announcement can earn the run-up without bearing the event jump (Gao et al., 2018a); see also Dubinsky et al. (2019).

Specification.

- *Signal:* an earnings calendar from `earnings_actuals`, under the labelled assumption that announcement dates are known at least 7 sessions ahead. The exit session X is the announcement date if the announcement is after the close, otherwise the session before it. Entry is at session $X - 5$.
- *Portfolio:* long one ATM straddle (strike nearest F from (3)) in the first expiry after the announcement. The premium budget is 0.5% of equity per event, and events may overlap.
- *Execution tier:* T2 at $\tau = 15{:}55$ on entry and exit (buy at the ask, sell at the bid).
- *Risk limit:* the loss is bounded by the premium.
- *Secondary:* the same trade held through the event.

Implementation. The engine is `wave2/cpp/h5_earnings.cpp`. The pre-registered rule filled 1 event, so the original primary is untestable. Amendments A1, A3 and A4 reach 75 fills. They choose the strike from quoted contracts, use a stock-mid reference when the parity forward is unavailable, and take quotes from 120 s windows. All were motivated by measured fill availability, and A4 is labelled exploratory.

Out-of-sample testing and results. The A4 variant lost 0.8% of premium per event in development ($t = -0.81$) and 6.3% in the holdout (11 events). Holding through the event was negative, consistent with the wave-0 EOD earnings study (Figure 13). The decision is reject.

6.8 H6: end-of-day momentum

Economic intuition and literature. Negative-gamma dealer hedging and late-informed trading may push the day's move further into the close (Gao et al., 2018b; Baltussen et al., 2021).

Specification.

- *Signal:* the return from the prior close to 15:55,

$$r = \frac{\text{mid}(15{:}55)}{\text{close}_{t-1}} - 1, \tag{8}$$

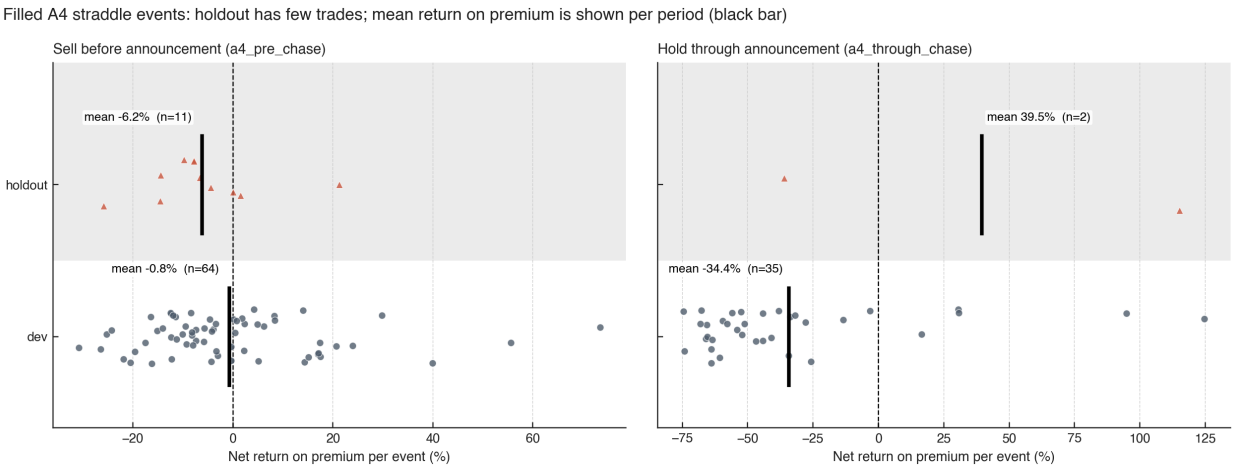


Figure 13: H5: event returns for the A4 variant (`wave2/figures/fig09_h5_events.png`).

Table 14: H6: end-of-day momentum, SPY and QQQ.

Sample	Ann. excess	Sharpe	NW t	Placebo percentile
Development	+0.8%	0.91	1.50	99.3
Holdout (2024H2)	−0.9%	−1.37		24

where close_{t-1} is the official closing auction print. The primary position is $\text{sign}(r)$ every day. The secondary trades only on days when the prior-day dealer gamma exposure is negative,

$$\text{GEX} = \sum_{\text{calls}} \Gamma \cdot \text{OI} \cdot 100 \cdot S^2 \cdot 0.01 \; - \; \sum_{\text{puts}} \Gamma \cdot \text{OI} \cdot 100 \cdot S^2 \cdot 0.01, \tag{9}$$

under the labelled assumption that dealers are long calls and short puts.

- *Portfolio:* 50% of equity in SPY and 50% in QQQ, each signed independently.
- *Execution tier:* entry at 15:55 plus latency against the equity NBBO (T2, size at most the displayed size) and exit at the closing auction print (T1).
- *Evaluation:* daily net return, Newey–West t, hit rate, and a placebo with randomized signs (1,000 draws).

Implementation. The engine is `wave2/cpp/h6_eodmom.cpp`. Amendment A2 responds to data availability: the equity quote window opens at 15:55:00, so the signal uses the mid of the first quote at 15:55:00.5, which is observable at that time. A missing equity-quote window blanked H6 in an early mechanics check, and the bug was fixed before any out-of-sample run. Displayed-size caps filled about 47% of the target notional.

Out-of-sample testing and results. The development Sharpe ratio is 0.91 ($t = 1.50$), at the 99.3rd percentile of 1,000 random-sign placebos. The holdout Sharpe ratio is −1.37, at the 24th percentile (Figure 14). The decision is reject.

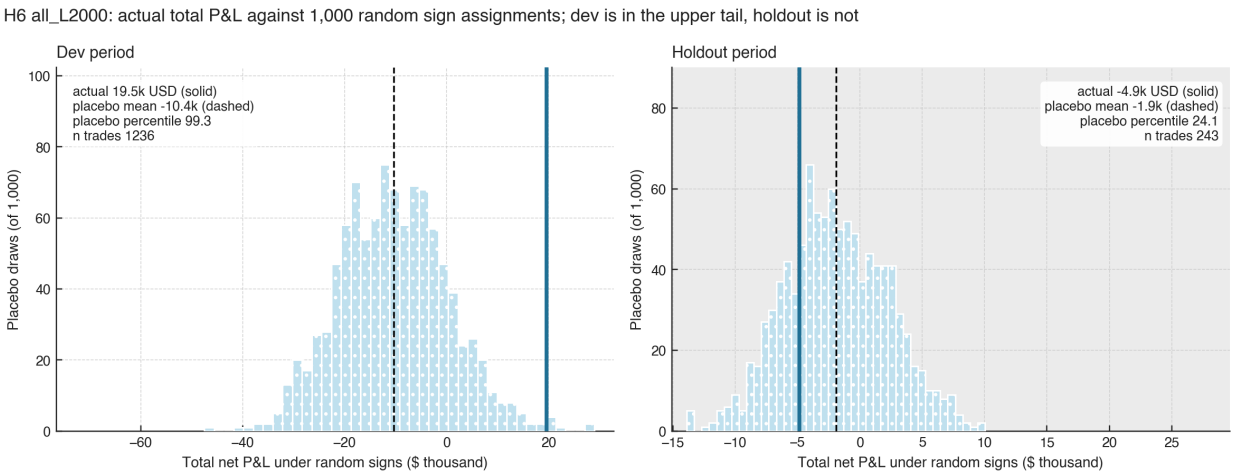


Figure 14: H6: actual results against random-sign placebos (`wave2/figures/fig10_h6_placebo.png`).

6.9 S1: pre-earnings announcement premium

Economic intuition and literature. Stocks earn higher returns in the days before scheduled earnings. This is consistent with compensation for announcement risk and with attention-driven buying pressure (Frazzini and Lamont, 2007; Barber et al., 2013).

Specification.

- *Signal:* the earnings calendar, under the assumption that dates are known at least 7 sessions ahead. X is the last session before the announcement (as in H5).
- *Rule:* buy at the close of $X - 5$ and sell at the close of X, before the announcement.
- *Hedge:* short SPY at the same auction, with the ratio set to the 60-session OLS beta of daily close-to-close returns ending the day before entry, capped to $[0.5, 2]$.
- *Portfolio:* 10% of equity per event, with overlapping events allowed.
- *Execution tier:* T1 closing prices.
- *Secondaries:* unhedged, and a 10-session window.
- *Test:* mean hedged event return (clustered by date) and the account Sharpe ratio.

Implementation. The engine is `wave3/cpp/s_events.cpp`, with official-close fills and a daily marked book. The official close equals the closing-cross print on 100% of tested days. The wave-3 stock book reconciles exactly to its ledger. Confirmatory split: development 2022-01-03 to 2023-12-29, test year 2024.

Out-of-sample testing and results. The development hedged event return is +0.46% (105 events, clustered $t = 1.47$), with annual excess return +2.4% (NW $t = 1.08$). The 2024 test is +0.06% per event (60 events, $t = 0.02$) and +0.4% annualized (Table 15). Unhedged, development is +1.10% per event ($t = 2.28$), mostly market exposure, and the 2024 test is about 0 (Figure 15). S1 survives the kill rule narrowly, but the premium did not appear in 2024. The result is not significant.

Table 15: S1: pre-earnings premium, beta-hedged.

Sample	Ann. excess	Sharpe	NW t	Per event	Events
Development	+2.4%	0.88	1.08	+0.46%	105
Test (2024)	+0.4%	0.17		+0.06%	60

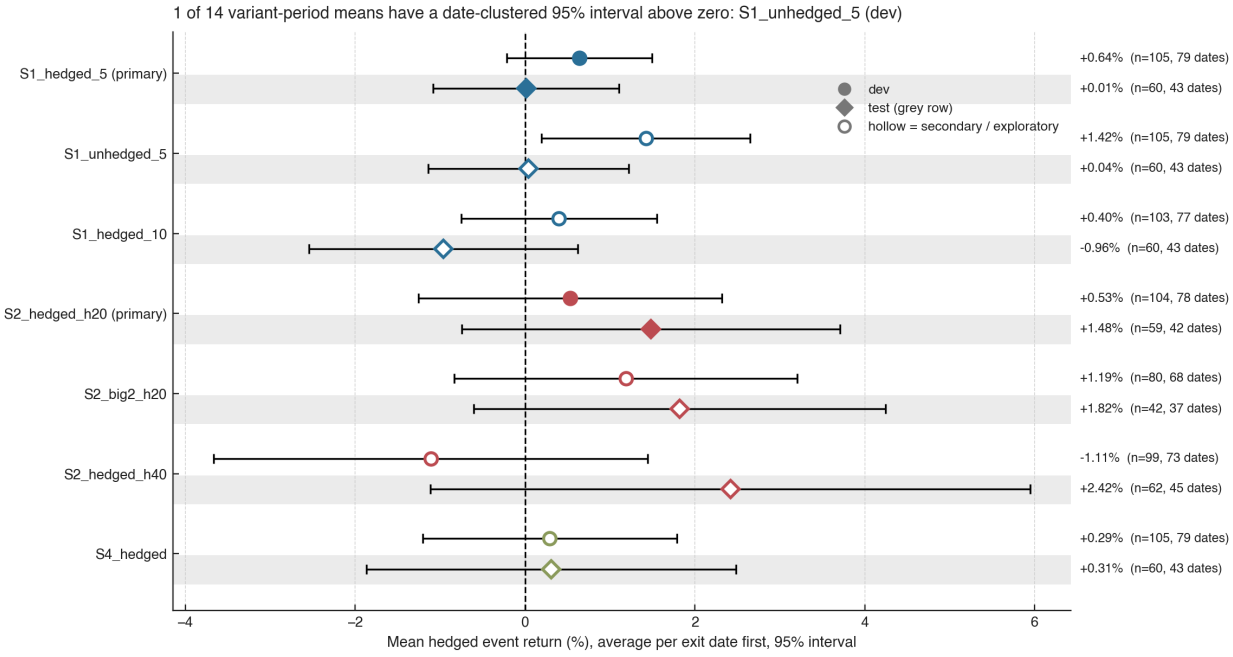


Figure 15: Wave-3 event strategies S1, S2 and S4 (`wave3/figures/fig13_w3_events.png`).

Table 16: S2: earnings-announcement-return drift, 20-session hold.

Sample	Ann. excess	Sharpe	NW t	Per event	Event t
Development	+3.4%	0.64	0.87	+0.50%	0.59
Test (2024)	+12.8%	1.63		+1.55%	1.32

6.10 S2: earnings-announcement-return drift

Economic intuition and literature. Prices may under-react to earnings news. The market's own reaction serves as a surprise measure that needs no analyst forecasts, which this dataset lacks (Brandt et al., 2008; Chan et al., 1996).

Specification.

- *Signal:* the earnings-announcement return, beta-adjusted,

$$\mathrm{EAR}_i = r_i - \beta_i \, r_{\mathrm{SPY}}, \tag{10}$$

  measured over the reaction window. For after-close announcers (AMC) the window runs from the close of announcement day A to the close of $A + 1$. For before-open announcers (BMO) it runs from the close of $A - 1$ to the close of A.
- *Rule:* take sign(EAR) at the next session's opening cross, hold 20 sessions and exit at the close, beta-hedged with SPY (60-session OLS beta, capped to $[0.5, 2]$).
- *Portfolio:* 10% of equity per event.
- *Execution tier:* T1. If the opening cross is missing, the order fills at that day's close.
- *Secondaries:* only events with $|\mathrm{EAR}| > 2\%$, and a 40-session hold.

Implementation. The engine is `wave3/cpp/s_events.cpp`, which is shared with S1 and S4. Confirmatory split: development 2022-01-03 to 2023-12-29, test year 2024.

Out-of-sample testing and results. Development gives +0.50% per event ($t = 0.59$), annual excess return +3.4% and Sharpe 0.64. The test gives +1.55% per event (59 events, $t = 1.32$), annual excess return +12.8% and Sharpe 1.63 (Table 16). The $|\mathrm{EAR}| > 2\%$ subset is similar. The 40-session hold was negative in development and positive in the test. S2 survives the kill rule but is not significant. The direction is consistent with the literature, but about 100 development events give low power.

6.11 S4: pre-announcement option flow

Economic intuition and literature. Informed trading may concentrate in options before scheduled news (Roll et al., 2010; Johnson and So, 2012). S4 asks whether the direction of option flow in the week before earnings predicts the announcement return.

Specification.

Table 17: S4: pre-announcement option flow (exploratory).

Sample	Ann. excess	Sharpe	NW t	Per event
Development	+2.2%	0.45	0.66	+0.41%
Test (2024)	+0.2%	0.05		+0.07%

- *Signal:* the delta-weighted flow ratio of (7), with numerator and denominator summed separately over sessions $X-5,\ldots,X-1$,

$$\text{Flow}^{\text{pre}} = \frac{\sum_{s=X-5}^{X-1} \text{num}_s}{\sum_{s=X-5}^{X-1} \text{den}_s}. \tag{11}$$

- *Rule:* buy or sell at the close of X in the direction of sign(Flow^{pre}). Exit at the first opening cross after the announcement, or at that day's close if the cross is missing. Beta-hedge with SPY.
- *Portfolio:* 10% of equity per event.
- *Status:* exploratory, because the signal is derived from H4.

Implementation. The engine is `wave3/cpp/s_events.cpp`. Because S4 is exploratory, it runs on the full 2022–2024 sample and cannot be confirmed here.

Out-of-sample testing and results. The result is +0.41% per event in development ($t = 0.38$) and +0.07% in 2024 (Table 17). We detect no effect at this sample size.

6.12 S5: SPXW premium held to settlement, timed by the VIX term structure

Economic intuition and literature. The variance risk premium is larger, and short-volatility exposure safer, when the VIX term structure is in contango (Carr and Wu, 2009; Bollerslev et al., 2009; Johnson, 2017). At 8–30 DTE, crossing the spread once costs about 0.6% of premium against about 8% for a 0DTE round trip (F1).

Specification.

- *Signal:* the primary filter trades when VIX3M/VIX > 1 on the entry date (EOD). Secondaries are unconditional entry and VIX $-$ RV21(SPX) > 0.
- *Rule:* every Friday, sell an SPXW iron fly at the EOD bid/ask. The short strike is nearest spot, and the wings sit at the strikes nearest $\pm 1.5\times$ the straddle mid. The expiry is the Friday expiry with DTE in $[26, 30]$ (otherwise the nearest in $[21, 35]$). Hold to PM settlement on `sp500_cash_level`.
- *Portfolio and risk:* maximum loss at most 1% of equity per position, with up to about 4 overlapping positions. The maximum loss per contract is

$$(w - \text{credit}) \times 100, \tag{12}$$

  with w the wing width.
- *Execution tier:* T3 entry, T0 exit.
- *Test:* account Sharpe, Newey–West t, and the alpha from a regression of daily returns on SPY daily returns, with beta reported.

Table 18: S5: 28-DTE SPXW iron fly, VIX-timed.

Sample	Ann. excess	Sharpe	NW t	Per trade (on max loss)
Development	-3.5%	-1.08	-1.37	-22.0% ($t = -2.99$)
Test (2024)	-8.9%	-1.59		-12.8% ($t = -1.27$)

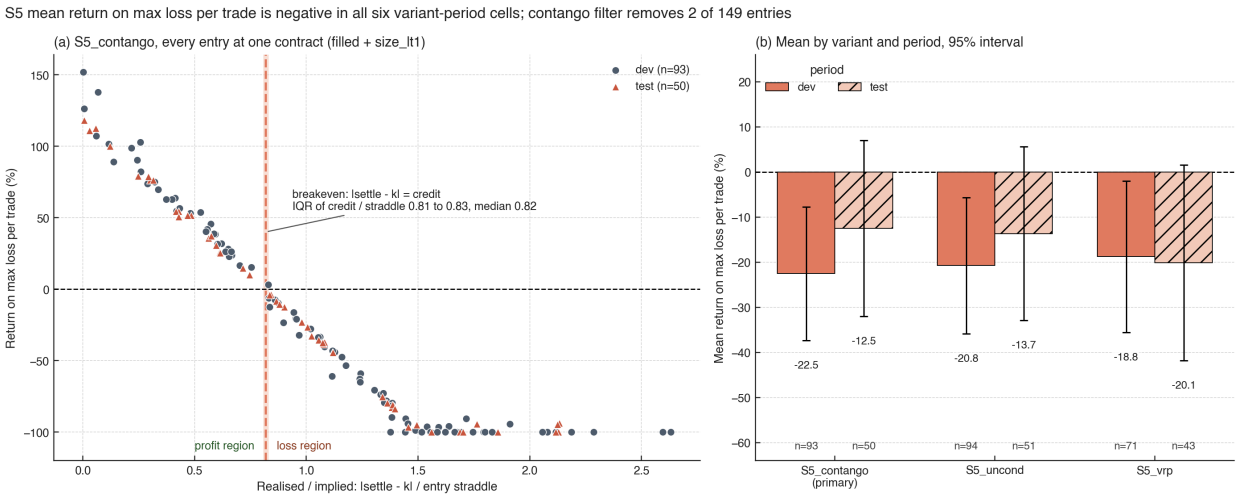


Figure 16: S5: realized settlement distance against the implied straddle (`wave3/figures/fig14_s5_vrp.png`).

Implementation. The engine is `wave3/cpp/s5_spxw_vix.cpp`, with T3 entry, T0 exit and daily EOD-mid marks. Amendment B1 was made after seeing development aggregate P&L and is disclosed. One contract often risks more than 1% of the account, and 64% of entries were skipped as `size_lt1`, concentrated in high-volatility weeks. The per-trade test therefore uses every entry at one lot. B1 changes the denominator of the per-trade test, not the trading rule, and the account-level series is reported as written. A hand-recomputed S5 trade matches the engine to the cent.

Out-of-sample testing and results. The development return on maximum loss is -22.0% per trade (93 trades, clustered $t = -2.99$), and 2024 is -12.8% ($t = -1.27$) (Table 18). Realized moves equalled the implied straddle: the mean $|\text{settle} - K|$ was 164 points against a mean straddle of 163. After buying wings, the fly loses (Figure 16). The contango filter bound on only 1 of about 95 development Fridays, so it could not time anything in this sample. The decision is reject.

6.13 S6: realized-minus-implied volatility straddles, exited by exercise

Economic intuition and literature. Options on stocks whose realized volatility exceeds implied volatility are cheap. Long-short straddle portfolios sorted on that gap earn positive returns in Goyal and Saretto (2009). Holding to expiry and exercising replaces the option exit spread with a stock auction trade.

Specification.

Table 19: S6: RV − IV straddles, long-short.

Sample	Ann. excess	Sharpe	NW t
Development	−2.6%	−0.73	−1.06
Test (2024)	−6.9%	−0.95	

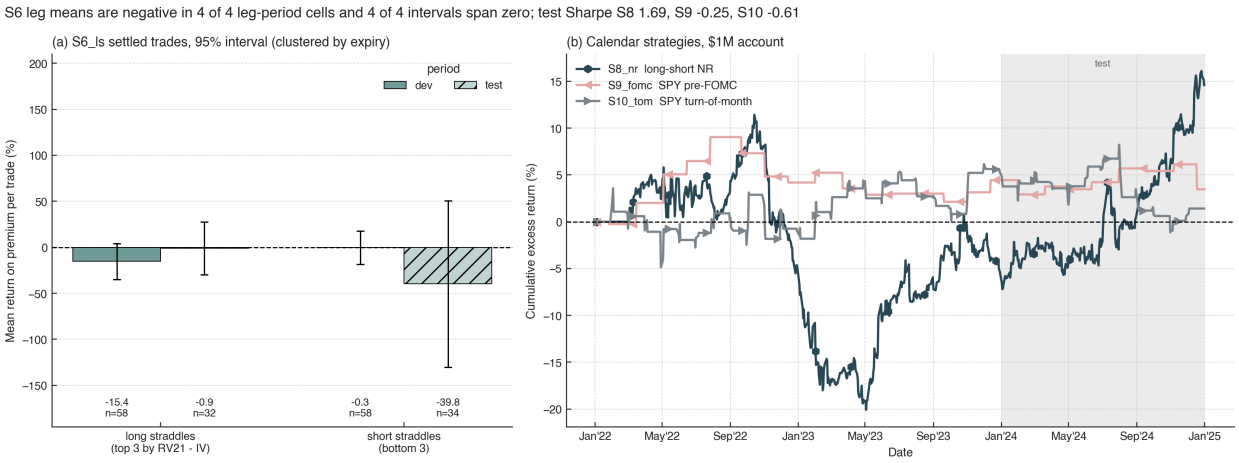


Figure 17: S6 and S8 results (`wave3/figures/fig17_s6_s8_calendar.png`).

- *Signal:* on each monthly expiry Friday, for each name,

$$\text{gap} = \text{RV21} - \text{IV}^{\text{ATM}}, \tag{13}$$

  where RV21 is computed from closes and IV^{ATM} is the mean IV of the call and put nearest spot in the next monthly expiry.
- *Rule:* buy ATM straddles on the 3 names with the largest gap and sell straddles on the 3 with the smallest. Enter at the EOD snapshot (buy ask, sell bid) and hold to expiry. At expiry, in-the-money legs are exercised or assigned at the official close, and the delivered shares are closed at the next opening cross (or the close if that is missing), with stock fees.
- *Portfolio:* premium-neutral, 2% of equity of premium per side, split equally across names. A secondary buys only the top 3.
- *Execution tier:* T3 entry, with exercise at the official close and a stock auction exit (T1).
- *Risk:* early assignment of short legs is not modelled, which is a known limitation.

Implementation. The engine is `wave3/cpp/s6_straddle_gs.cpp`. With three names per side, the sort cannot reproduce the decile spreads of the original study.

Out-of-sample testing and results. The long leg lost 13.7% of premium per trade in development, and the short leg was about 0. The long-short annual excess return is −2.6% in development (Sharpe −0.73, NW $t = -1.06$) and −6.9% in 2024 (Sharpe −0.95) (Table 19; Figure 17). The decision is reject.

Table 20: S7: pin fly against ATM fly (exploratory).

Split	Ann. excess	Sharpe	NW t
Development	+1.0%	0.28	0.38
Test (2024)	−0.1%	−0.03	

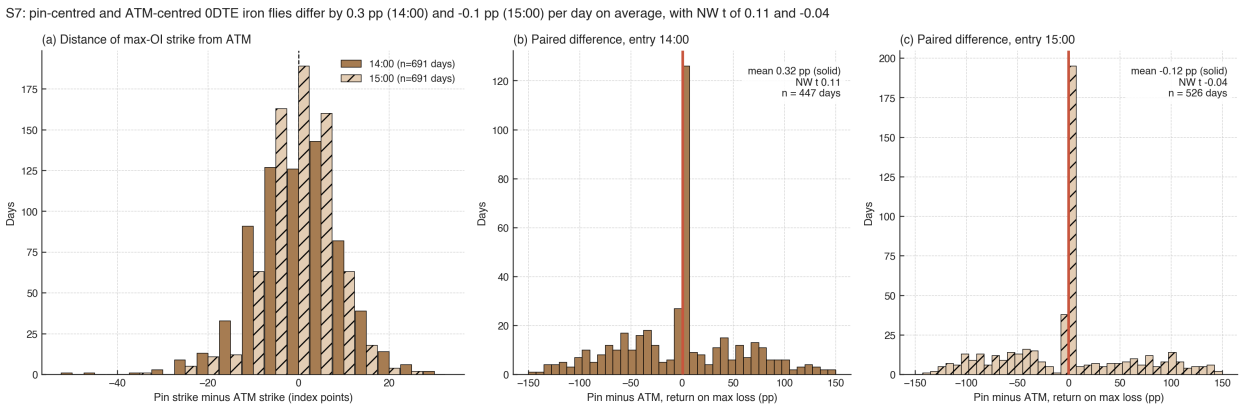


Figure 18: S7: pin strike against ATM strike (`wave3/figures/fig16_s7_pin.png`).

6.14 S7: 0DTE pin at the maximum-OI strike

Economic intuition and literature. Hedging of large open interest may pull prices toward heavily traded strikes at expiry (Ni et al., 2005). S7 asks whether centering an iron fly on the maximum-OI strike beats centering it on the ATM strike.

Specification.

- *Rule:* H1's iron fly at 14:00 and 15:00, centered on the strike with the largest call plus put OI from the last record before expiry, among strikes within ±1 straddle of F from (3). The comparison is the ATM fly on the same days.
- *Execution tier:* T2 entry, T0 settlement, realistic fees.
- *Status:* exploratory, because the design is derived from H1. There is no primary.
- *Evaluation:* the pin-minus-ATM difference in return on risk, over paired days.

Implementation. The engine is `wave3/cpp/s7_pin.cpp`. It runs on the full 2022–2024 sample, so the development and test columns below are splits of one exploratory sample. S7 cannot be confirmed here.

Out-of-sample testing and results. Pin minus ATM is +0.3 percentage points of return on risk at 14:00 (447 paired days, $t = 0.11$) and −0.1 points at 15:00 ($t = -0.04$). The pin strike equalled the ATM strike on 18% (14:00) and 27% (15:00) of days, and its median distance from ATM was zero (Figure 18). We detect no pin effect.

Table 21: S8: daytime-reversal frequency, long-short terciles.

Sample	Ann. excess	Sharpe	NW t
Development	−1.8%	−0.13	−0.20
Test (2024)	+19.6%	1.69	

6.15 S8: overnight returns and daytime reversals

Economic intuition and literature. Overnight noise-trader demand may push prices up overnight, and intraday arbitrageurs may push back. Akbas et al. (2022) relate the monthly frequency of negative daytime reversals (positive overnight return followed by a negative intraday return) to future returns. We pre-registered a positive sign (high frequency, higher next-month return), following that paper. The citation is to be verified against the source.

Specification.

- *Signal:* NR is the fraction of the last 21 sessions with an overnight return above 0 and an intraday return below 0, computed from auction prints,

$$\mathrm{NR}_i = \frac{1}{21}\sum_{s=1}^{21} \mathbf{1}\{r_{i,s}^{\mathrm{ovn}} > 0,\ r_{i,s}^{\mathrm{intra}} < 0\}. \tag{14}$$

- *Universe:* names with at least 15 valid sessions.
- *Portfolio:* at the month-end close, go long the top third and short the bottom third by NR. The book is dollar neutral and held for one month.
- *Execution tier:* T1 closes.
- *Status:* confirmatory but partially informed, because S8 is closest to wave-2 H3.

Implementation. The engine is `wave3/cpp/s_calendar.cpp`. Split-adjusted share carry was fixed before the run. With 12–13 eligible names per month, each tercile holds four names, so the ranking is coarse.

Out-of-sample testing and results. Development annual excess return is −1.8% (Sharpe −0.13, NW $t = -0.20$) and the 2024 result is +19.6% (Sharpe 1.69) (Table 21). The primary is rejected by the development criterion of the kill rule, so the 2024 result does not rescue it.

6.16 S9 and S10: FOMC drift and turn-of-the-month

Economic intuition and literature. US equities may earn much of their excess return in the 24 hours before scheduled FOMC announcements (Lucca and Moench, 2015) and around month turns (Ariel, 1987; Lakonishok and Smidt, 1988). The month-turn effect is consistent with payment-flow timing.

Specification.

- **S9 (FOMC):** buy SPY at the close of the session before the announcement and sell at the close of announcement day. The data has no 14:00 price, so the post-announcement reaction is included (a known dilution). The portfolio is 100% of equity over 24 events.

Table 22: S9 and S10: calendar strategies on SPY.

Strategy	Sample	Ann. excess	Sharpe	NW t
S9	Development	+2.3%	0.53	0.74
S9	Test (2024)	−0.9%	−0.25	
S10	Development	+3.0%	0.36	0.55
S10	Test (2024)	−3.6%	−0.61	

- **S10 (turn of the month):** buy SPY at the close of the second-to-last session of the month and sell at the close of the third session of the next month. The portfolio is 100% of equity.
- *Execution tier:* T1 official closes for both. Both are confirmatory but underpowered.

Implementation. The engine is `wave3/cpp/s_calendar.cpp`, with official-close fills. S9 has 24 events in total, so it cannot detect anything smaller than a very large effect.

Out-of-sample testing and results. Both strategies are positive in development (S9 +2.3% per year, Sharpe 0.53; S10 +3.0%, Sharpe 0.36) and negative in 2024 (Table 22). Both are rejected.

7 Capital allocation across strategies

7.1 Motivation and setup

Most strategies in this study commit only part of their capital. The option sleeves cap risk at 0.5–1% of equity per position, and the event sleeves hold positions on a fraction of days. A combined book therefore needs an explicit rule for capital that is not deployed. We park it in the Secured Overnight Financing Rate (SOFR). We then ask whether a portfolio of sleeves, with weights fixed from 2022–2023 data alone, would have added return over SOFR in 2024.

Let $x_{i,t}$ be the daily excess return of sleeve i over cash. These are the series the C++ engines write, one per pre-registered primary; the 13 sleeves are listed in Table **??**. Let c_t be the SOFR accrual from session $t-1$ to t, using the fixing of session $t-1$ and an actual/360 day count. The New York Fed publishes the series; we downloaded it from its public API (770 fixings, 2021-12-01 to 2024-12-31). The portfolio total return is

$$r_t^p \;=\; c_t \;+\; \sum_{i=1}^{13} w_i\, x_{i,t}, \qquad w_i \ge 0, \quad \sum_i w_i \le 1. \tag{15}$$

A weight w_i is the share of capital run through sleeve i. Inside a sleeve, idle cash already earns the cash rate, so (15) credits SOFR on all undeployed capital. The sleeves' own excess returns are measured against the 1-month T-bill. We treat the small T-bill/SOFR basis as negligible, which is an approximation.

7.2 Allocation rules

All rules are long-only. Covariances use the Ledoit–Wolf shrinkage estimator toward a scaled identity (Ledoit and Wolf, 2004).

1. **1/N**: equal capital to every sleeve. This is the benchmark that estimated rules often fail to beat out of sample (DeMiguel et al., 2009).
2. **Inverse volatility**: $w_i \propto 1/\hat{\sigma}_i$, normalized to sum to one.
3. **Minimum variance**: $\min_w w^\top \hat{\Sigma} w$ subject to $\sum_i w_i = 1$.
4. **Markowitz (Bayes–Stein, 25% cap)**: maximize $w^\top \tilde{\mu} / \sqrt{w^\top \hat{\Sigma} w}$ subject to $0 \le w_i \le 0.25$ and $\sum_i w_i = 1$ (the Sharpe ratio is scale-free, so the budget is fixed inside the optimizer) (Markowitz, 1952). Here $\tilde{\mu}$ shrinks sample means toward the minimum-variance portfolio mean with Jorion's intensity (Jorion, 1986).
5. **Development-screened 1/N**: equal weight on sleeves whose estimation-window Newey–West t exceeds 1.

Schedules. Weights are estimated on 2022-01-03 to 2023-12-29 and applied two ways:

- **Static**: estimated once and held through 2024.
- **Walk-forward**: re-estimated at each 2024 month start on an expanding window.

Volatility target. We also scale the static 1/N and Markowitz books to a 5% in-sample volatility, with leverage capped at 3, following the logic of volatility targeting (Moreira and Muir, 2017). Leverage multiplies excess returns, and financing is implicitly at SOFR.

7.3 Results

Table 23 reports each rule. In 2024, holding SOFR alone returned 5.40% (mean annualized accrual 5.26%).

Table 23: Allocation results. Excess and Sharpe are relative to SOFR. Weights never use 2024 data, except that the walk-forward schedule uses 2024 data up to each rebalance.

Rule	Schedule	Total	Excess	Vol	Sharpe	Max DD
2022–2023, in-sample						
1/N	static	3.93%	0.50%	2.03%	0.25	−1.34%
Inverse vol	static	3.51%	0.08%	0.76%	0.11	−0.42%
Minimum variance	static	3.33%	−0.10%	0.42%	−0.23	−0.30%
Markowitz (B–S, cap 25%)	static	6.22%	2.67%	1.65%	1.62	−0.86%
Dev-screened 1/N	static	7.42%	3.84%	3.10%	1.24	−1.93%
2024, out-of-sample						
1/N	static	9.91%	4.21%	2.10%	2.00	−0.89%
Inverse vol	static	5.91%	0.49%	1.01%	0.48	−0.45%
Inverse vol	walk-forward	5.99%	0.57%	1.00%	0.56	−0.43%
Minimum variance	static	5.16%	−0.23%	0.55%	−0.41	−0.23%
Minimum variance	walk-forward	5.12%	−0.27%	0.51%	−0.53	−0.22%
Markowitz (B–S, cap 25%)	static	9.31%	3.67%	1.94%	1.89	−0.74%
Markowitz (B–S, cap 25%)	walk-forward	10.43%	4.69%	2.27%	2.07	−0.89%
Dev-screened 1/N	static	10.05%	4.35%	2.67%	1.63	−1.13%
Dev-screened 1/N	walk-forward	14.01%	7.91%	3.54%	2.23	−1.13%
1/N, 5% vol target (×2.46)	static	16.75%	10.37%	5.18%	2.00	−2.29%
Markowitz, 5% vol target (×3.00)	static	17.45%	11.00%	5.82%	1.89	−2.61%

Figures 19–21 show the weights, the sleeve correlations (median off-diagonal −0.01, largest absolute value 0.25) and the 2024 paths.

Reading the table.

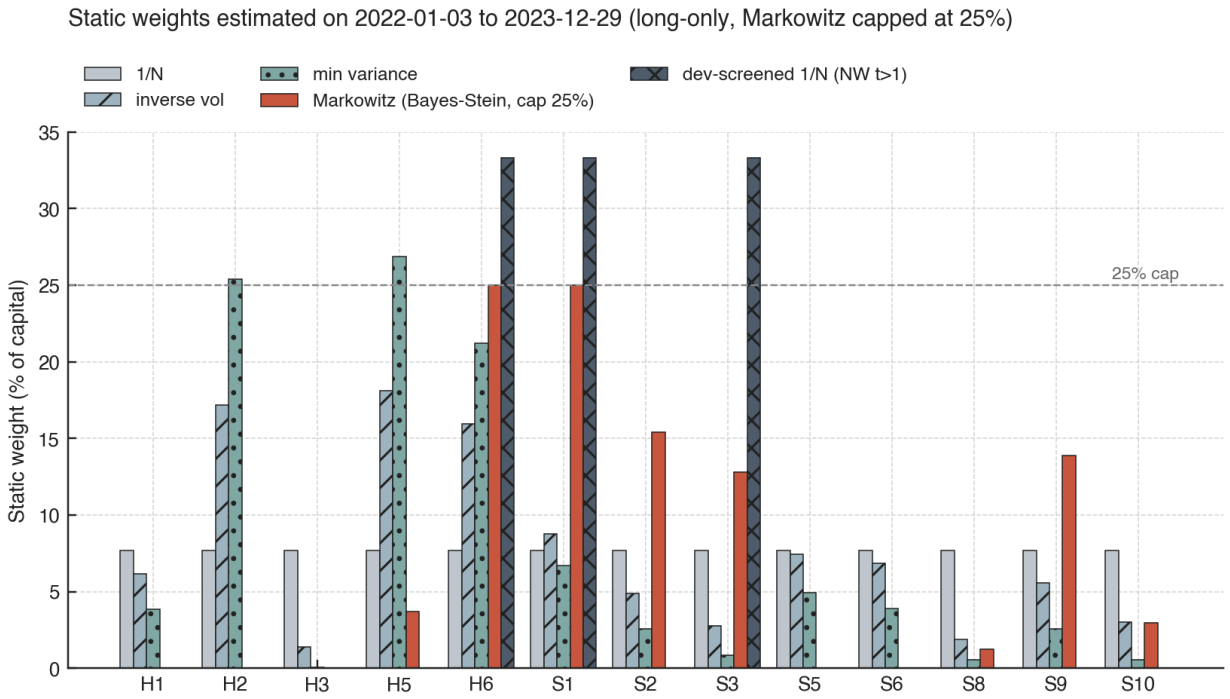


Figure 19: Static weights estimated on 2022–2023 for each allocation rule. The dashed line marks the 25% Markowitz cap.

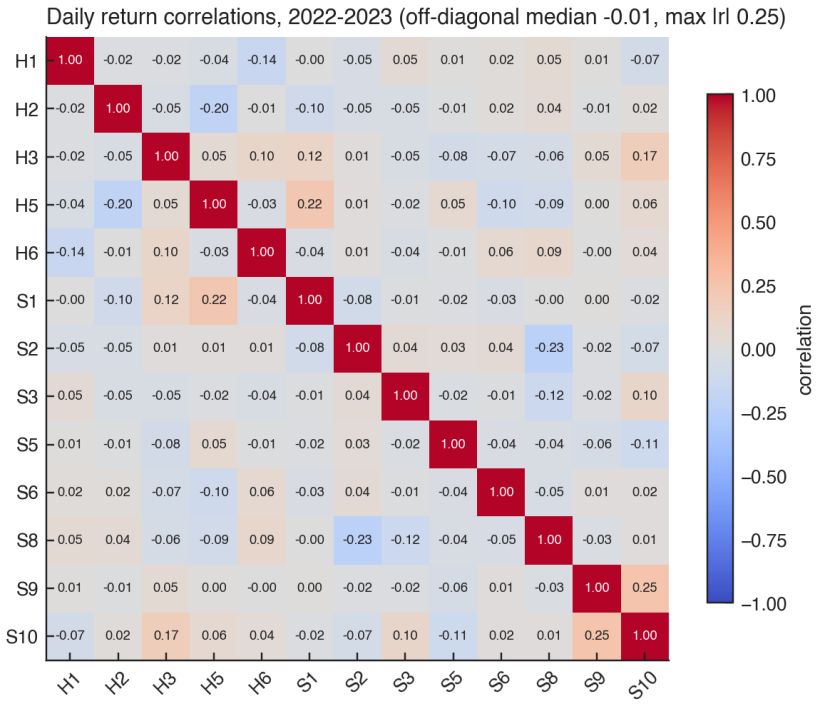


Figure 20: Correlations of daily sleeve returns, 2022–2023.

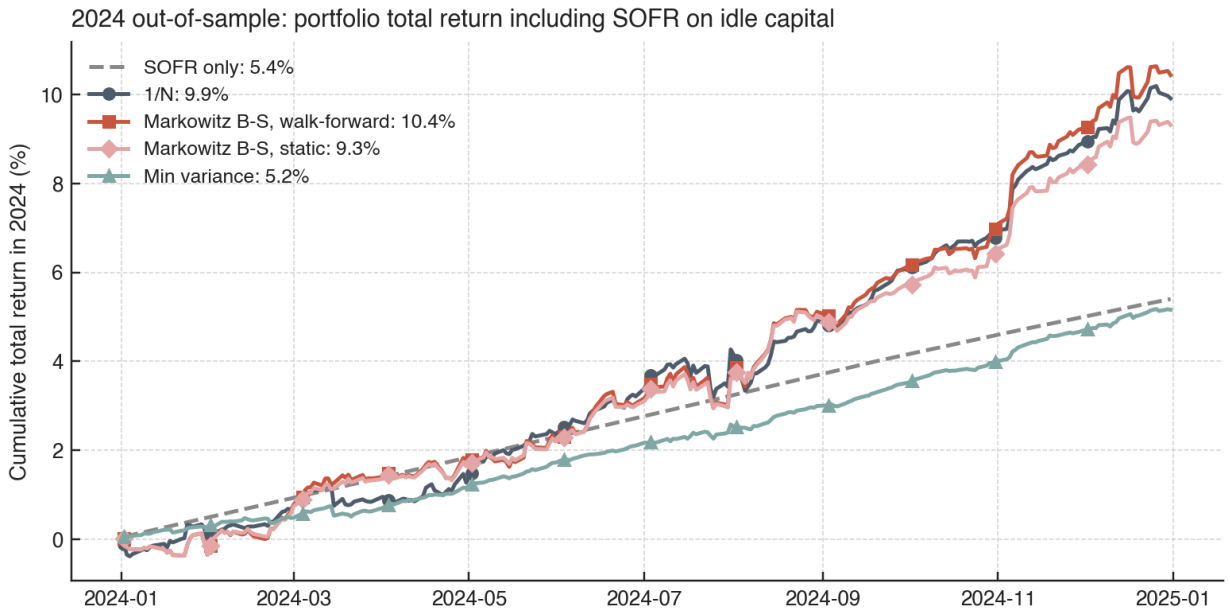


Figure 21: Cumulative 2024 total return of selected allocation rules, including SOFR on idle capital.

- **Minimum variance.** It loads on the sleeves with the lowest volatility: the small-sized overnight and pre-earnings straddle books (H2, H5), which carry 0.5–0.6% annual volatility at their risk caps (2022–2023). It returns roughly SOFR. Low volatility here reflects small position sizes, not diversification.
- **Markowitz with shrunk means.** It avoids that trap. It puts its weight on the sleeves with positive estimation-window means. H6 and S1 reach the 25% cap; S2, S3 and S9 receive 13–15%; H5, S8 and S10 receive at most 4%. It earns 3.7–4.7% over SOFR in 2024.
- **1/N.** It earns 4.2% over SOFR. That return comes largely from sleeves that were weak in 2022–2023 and strong in 2024 (H3 overnight equity, S8): equal weighting kept them in the book.
- **Uncertainty.** No 2024 Sharpe ratio in Table 23 rests on more than 252 daily observations. The standard error of a one-year Sharpe estimate is about 1, so the differences between the 1/N, Markowitz and screened rules are not statistically distinguishable.

7.4 Why this 2024 test is weaker than it looks

Three caveats limit what Table 23 can support.

1. **Known sleeve outcomes.** No weight uses 2024 returns, but the sleeves' 2024 outcomes were already known when this allocation study was designed: the wave-2 holdout and the wave-3 test both cover 2024. The set of sleeves is "all primaries", which limits but does not remove that knowledge. The option-flow sleeve uses the 1-day hold (the wave-2 H4 specification with realistic fees), not the 5-day hold designated in the wave-3 pre-registration; the estimation window would have chosen it as well (Sharpe 0.95 against −0.46).
2. **Fees and contract sizes.** Wave-2 sleeves (H1–H6) carry the wave-2 fee assumptions, which understate single-name and ETF option fees by about \$0.43 per contract. Scaling a sleeve by w_i also ignores integer contract sizes. At $w_i = 0.077$ on a \$1M account, several option sleeves would trade fractional contracts.
3. **Leverage.** The volatility-targeted books use 2.46–3 times leverage on sleeves whose risk is

dominated by rare tail days (H1, S5). Their 2024 Sharpe ratios are the unlevered ratios. Their drawdowns would scale with any tail event that 2024 did not contain.

7.5 Allocation proposal

These results suggest a combined book can add return over SOFR with low volatility, but they do not establish it. We propose a staged allocation in which every stage has a pre-registered gate.

1. **Default: SOFR.** All capital not assigned to a sleeve earns SOFR, through overnight reverse repo or a government money-market fund. This is the benchmark every stage must beat after costs.
2. **Stage 1, paper portfolio in 2025.**
   - Freeze the Bayes–Stein Markowitz rule (25% cap, monthly walk-forward on an expanding window) and the 1/N rule as a control.
   - Universe: the 13 sleeves, with wave-2 sleeves re-costed at the wave-3 fee schedule.
   - Run both on 2025 data without changes.
   - Pass gate: positive excess over SOFR, a Sharpe above that of the 1/N control, and a drawdown below 2% unlevered.
3. **Stage 2, small live risk budget.** Only if Stage 1 passes, allocate to the two sleeves with replicated evidence (S3 option flow and S2 announcement drift) at a combined 5% annualized volatility budget, with the remainder in SOFR. Each sleeve has a kill switch: stop if its live rolling 63-session Sharpe is below -1, or if its drawdown exceeds twice its 2022–2024 maximum.
4. **Excluded sleeves.** The short-volatility sleeves (H1, S5) and the straddle sleeves (H2, H5, S6) have negative estimation-window means and receive zero Markowitz weight. They stay out unless a new pre-registered test supports them.

This proposal describes a research protocol for the strategies in this study. It is not investment advice. The allocation code is `alloc/allocate.py`; weights, sleeve statistics and daily portfolio returns are in `alloc/`.

8 What we learned

1. **Execution cost decides most of these strategies.** Spread crossing turned the documented overnight option premium negative (H2): it alone was about four times that premium. It is also why waves 0 and 1 failed. Designs that cross once and settle (H1, S5), or trade only in auctions (H3, H4, S1–S3, S8–S10), remove most of that cost. The strategies that remained were then limited by their signals, not their costs.
2. **Short-volatility premia were not available in 2022–2024 at these strikes.** Realized moves matched implied straddles for 28-DTE SPXW (S5), and 0DTE flies were negative or flat (H1, S7). This period contains a bear market and a strong recovery, so the result may be period-specific.
3. **Information in option flow is the most persistent signal.** It appeared in the wave-2 development and holdout periods and survived realistic fees in the wave-3 test. It decays within days. Its development t-statistics stay below 2, and multiplicity-adjusted evidence is weak.
4. **The data limits which questions can be asked.** Trade-sampled quotes make straddle-level intraday tests (H2, H5) depend on amendments, and 15 names limit every cross-sectional test.

9 Limitations and next steps

Out-of-sample periods. The wave-3 test year (2024) was not a virgin holdout: wave-2 results on 2024 had been seen, though none of the wave-3 signals had been computed on it. S3, S4 and S7 are exploratory and need 2025 data.

Unmodelled execution and lifecycle. The option tape's sampling scheme is undocumented, and fill rates may differ in live markets. Early assignment (S6), the SPX AM settlement value (F2) and complex-order books are not modelled. Fees follow a public schedule, not executed broker statements.

Next steps.

- Freeze `wave3/cpp/s3_flow.cpp` (1-day hold, all-flow) and `wave2/cpp/h4_info.cpp` as they are and run each once on 2025 data. Pre-register the decision rule: Fama–MacBeth $t > 2$ and positive net Sharpe.
- Obtain a quote-event option feed (not trade-attached) to make H2 and H5 testable as originally written.
- Extend H4 and S3 to at least 100 names. Breadth is the binding constraint.

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